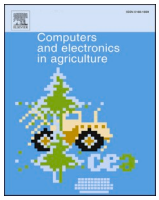
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Deep learning for plant stress detection: A comprehensive review of technologies, challenges, and future directions

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ABSTRACT

Deep learning (DL)-based systems have emerged as powerful methods for the diagnosis and treatment of plant stress, offering high accuracy and efficiency in analyzing imagery data. This review paper aims to present a thorough overview of the state-of-the-art DL technologies for plant stress detection. For this purpose, a systematic literature review was conducted to identify relevant articles for highlighting the technologies and approaches currently employed in the development of a DL-based plant stress detection system, specifically the advancement of image-based data collection systems, image preprocessing techniques, and deep learning algorithms and their applications in plant stress classification, disease detection, and segmentation tasks. Additionally, this review emphasizes the challenges and future directions in collecting and preprocessing image data, model development, and deployment in real-world agricultural settings. Some of the key findings from this review paper are: *Training data:* (i) Most plant stress detection models have been trained on Red Green Blue (RGB) images; (ii) Data augmentation can increase both the quantity and variation of training data; (iii) Handling multimodal inputs (e.g., image, temperature, humidity) allows the model to leverage information from diverse sources, which can improve prediction accuracy; *Model Design and Efficiency:* (i) Self-supervised learning (SSL) and Few-shot learning (FSL)-based methods may be better than transfer learning (TL)-based models for classifying plant stress when the number of labeled training images are scarce; (ii) Custom designed DL architectures for a specific stress and plant type can have better performance than the state-of-the-art DL architectures in terms of efficiency, overfitting, and accuracy; (iii) The multi-task learning DL structure reuses most of the network architecture while performing multiple tasks (e.g., estimate stress type and severity) simultaneously, which makes the learning much faster; *Application and Practicality:* (i) Training data collected from different growth settings or environmental conditions is important to increase the generalizability of the DL model; (ii) Overlapping stress symptoms can confuse the DL model; overlapping stress symptoms can be identified as a separate label such as "others" to solve this problem; (iii) Plant stress detection apps should have offline accessibility because remote field areas may not have internet access. This review can help shape the development of novel approaches, strategies, and applications that address the present gaps and challenges in plant stress detection.

Abbreviations: AI, Artificial Intelligence; AWSR, Accuracy of water stress recognition; AWTR, Accuracy of three water treatments recognition; CAE, Convolutional autoencoders; C-Drop, Clustering-based drop; CNN, Convolutional Neural Network; CWSI, Crop Water Stress Index; DCNN, Deep convolutional neural network; DL, Deep learning; ELISA, Enzyme-linked immunosorbent assay; FCN, Fully convolutional neural network; FSL, Few shot learning; GAN, Generative adversarial network; IDC, Iron deficiency chlorosis; IoU, Intersection over Union; ITK-net, "image-text" multi-modal collaborative representation and knowledge assistance network; LSTM, Long short-term memory; MCE, Multiscale conformer encoder; ML, Machine learning; MTL, Multi-task learning; NasNet, Neural Architecture Search Network; NDVI, Normalized difference vegetation index; NearIR, Near-infrared; PRISMA, Preferred reporting items for systematic reviews and *meta*-analyses; PSPNet, Pyramid Scene Parsing Network; RCNN, Region-Based Convolutional Neural Network; ResNet, Residual network; RGB, Red Green Blue; SMOTE, Synthetic minority oversampling technique; sRN, Simple recurrent network; SSD, Single shot detector; SSL, Self-supervised learning; SVM, Support vector machine; TL, Transfer learning; UAV, Unmanned aerial vehicle; UGV, Unmanned ground vehicle; VGG, Visual geometry group; ViT, Vision Transformer; YOLO, You Only Look Once.

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1. Introduction

In recent years, image-based deep learning models (DL) have demonstrated remarkable success in precision agriculture, including yield prediction, phenotyping, weed detection, soil nutrient and field condition monitoring, irrigation management, and plant stress detection. Plant stress is one of the greatest hazards to crops, significantly reducing both quality and production (Cattivelli et al., 2008). Therefore, rapid and accurate diagnosis of plant stress is urgently needed in crop measurement (Gao et al., 2020b). DL has emerged as a powerful tool for plant stress detection and monitoring through its advanced capabilities for analyzing and interpreting complex patterns in plant images and spectral data. DL-based stress detection systems can support farmers in making prompt, well-informed decisions that can significantly enhance the profitability, quality, and health of their crops. Additionally, DL can reduce the subjectiveness in measures of phenotypic responses of plants to treatments and thus improve scientific analyses.

The selection of DL architecture depends on input data, such as the varied complexities of visual symptoms exhibited by different plant stressor types. Some plant stressors have spatially localized effects, such as localized pest damage or disease lesions, while others affect the entire plant or field uniformly, such as drought stress. Plant stressors can be categorized into biotic and abiotic factors (Fig. 1). Biotic stress factors include pathogens such as fungi, bacteria, viruses, and pests that infect plants, leading to a wide range of symptoms, including discoloration, wilting, and even plant death (Georgieva and Vassileva, 2023). Abiotic stress encompasses environmental conditions that limit plant growth, such as extreme temperature, drought, flooding, salinity, metal, and nutrients (Y. Zhang et al., 2023). These stressors can adversely affect plant metabolism, physiological processes, and overall growth, leading to reduced yield and crop losses. It is crucial to detect these stressors to maximize the agricultural productivity and minimize resource use. Several studies have utilized image-based DL systems in detecting both biotic and abiotic stresses, such as fungal and bacterial diseases (Venal et al., 2020), iron deficiency (Li et al., 2020), drought stress (Dao et al., 2021a), nitrogen deficiency (Azimi et al., 2021a), salt stress (Khotimah et al., 2022), herbicide damage (Gómez-Zamanillo et al., 2023), potassium deficiency (Ghosal et al., 2018), light stress (Hao et al., 2020), and chemical injury (Venal et al., 2019). These DL-based models have been proven better than the traditional methods for plant stress detection in terms of efficiency and scalability.

Farmers and agronomists often rely on visual observation to identify signs of stress, such as wilting, discoloration, or abnormal growth patterns. However, this method is admittedly subjective and error-prone (Ghosal et al., 2018) and requires specialized knowledge and/or considerable manpower (Chen et al., 2023). Unlike subjective human observation, DL models provide objective and consistent analysis of plant stress across large areas, reducing human error and variability in detection and leading to more reliable results. Another traditional approach for plant stress detection is testing soil and plant tissue samples for nutrient content and other chemical properties. Although these tests provide valuable information, they are destructive methods, and they can be time-consuming and may not provide real-time data, whereas DL models are faster and can provide real-time diagnosis. Another traditional method is chlorophyll fluorescence measurement (Kalaji et al., 2023), which can assess the efficiency of photosynthesis and identify stress related to light absorption and energy transfer. However, this process often requires controlled environments. Molecular-based approaches, such as ELISA are time-consuming, labor-intensive, and not suitable for rapid, large-scale screening and monitoring (Galieni et al., 2021); in contrast, DL models can analyze large volumes of image data quickly, enabling rapid screening and monitoring of plant stress across extensive agricultural fields.

Imagery data is widely utilized in plant stress detection due to its ability to represent plant stress symptoms, like discoloration, wilting, and necrosis. Moreover, collecting data using imaging sensors (e.g., RGB, multispectral, hyperspectral) that are mounted on machines (i.e., unmanned ground vehicles [UGV], unmanned aerial vehicles [UAV], satellites, and airplanes) is non-destructive and applicable over large geographical areas (Shin et al., 2021a). The development of DL methods has revolutionized image analysis tasks. DL models provide high performance in computer vision tasks, such as classification, object

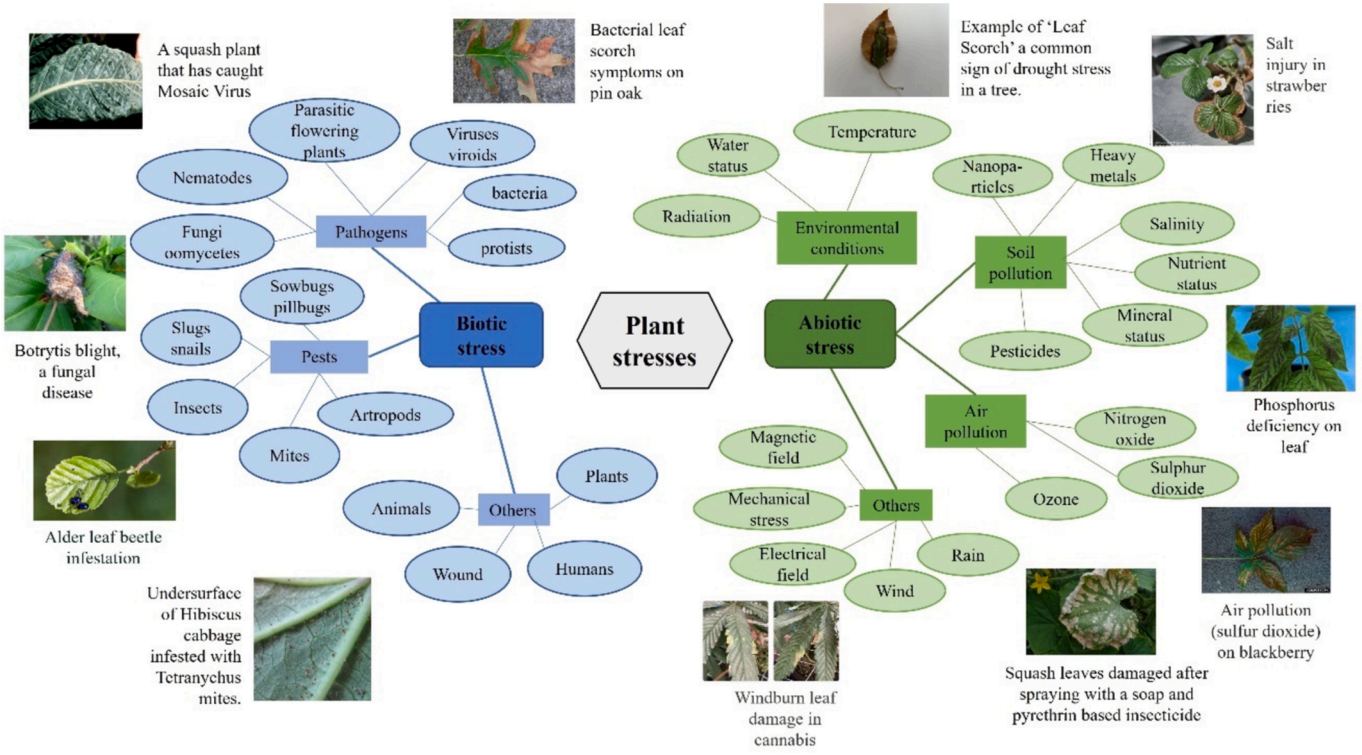


Fig. 1. Illustration of biotic and abiotic plant stressors.

detection, segmentation, and augmentation. The accessibility of modern deep learning frameworks such as TensorFlow and PyTorch have made it easier for practitioners to implement, train, and deploy deep learning models, leading to widespread adoption in various fields. There are several DL architectures being used in plant stress detection tasks, such as VGG16, ResNet, Inception, EfficientNet, Xception, ShuffleNet, and MobileNet.

The objective of this paper is to summarize the current state-of-the-art in different stages of developing a plant stress detection system, including image acquisition, preprocessing and deep learning methods. This review discusses emerging trends, common methodologies, and recurring patterns, which can help researchers and practitioners to understand the prevailing approaches and areas of focus within the field. Additionally, it assesses the effectiveness of various deep learning models and techniques, which can be useful in determining the strengths, weaknesses, and potential areas for development of current methodologies. Furthermore, it identifies challenges and gaps in current approaches and proposes future directions for potential solutions of these issues. This guidance can inform the development of new methodologies, techniques, and applications to address current gaps and challenges.

The remaining sections of this survey are organized as follows. Section 2 includes the method of this paper which explains how literature was retrieved and selected to review in a systematic manner. Section 3 contains the results, and describes the image acquisition technologies, preprocessing and deep learning methods used in recent studies. In Section 4, the challenges in plant stress detection system are described. In Section 5, the future directions for development are discussed. Finally, the paper is briefly summarized in Section 6.

2. Materials and methods

The primary goal was to identify articles relevant to the deep learning and computer vision- based plant stress detection system. For this purpose, a systematic literature review was undertaken across three prominent academic databases: ScienceDirect, Scopus, and Web of Science as depicted in Fig. 2. The search strategy comprised a combination of keywords ("Plant stress" OR "Crop stress" OR "Biotic" OR "Abiotic") AND "Deep learning" AND "Image" to retrieve the targeted literatures. Also, the retrieval process had the condition of selecting articles between the years 2014–2024. Following this initial retrieval stage, the abstracts of the obtained articles underwent manual examination, and the irrelevant articles were excluded from further analysis. Furthermore, duplicate entries were removed from the final compilation of selected articles. Following the selection of relevant articles, each stage of the plant stress detection system—image acquisition, image preprocessing, and deep learning technologies—was thoroughly described for this review paper.

3. Results

3.1. Outcome of the systematic search and selection process

During the literature retrieval stage, 1645 articles were found using the particular keywords. Among them, ScienceDirect had the highest amounts of articles (1345), followed by Scopus (162) and Web of Science (138). However, not all of them were relevant to our interest, as their research was related to plant phenotyping, using conventional ML algorithms, or non-imagery data, and such articles, review papers, and book chapters were discarded. There were some duplicate articles across the three databases, which were also removed to provide a unique list of articles. After removing irrelevant and duplicate articles, 163 unique articles remained, with 94 from ScienceDirect, 53 from Scopus, and 16 from Web of Science.

A trend of publications in the field of plant stress detection utilizing deep learning and imaging technologies is illustrated in Fig. 3. It demonstrates that the first article was published in Scopus in 2017 (Fig. 3a). ScienceDirect, however, has the most articles published in this field.

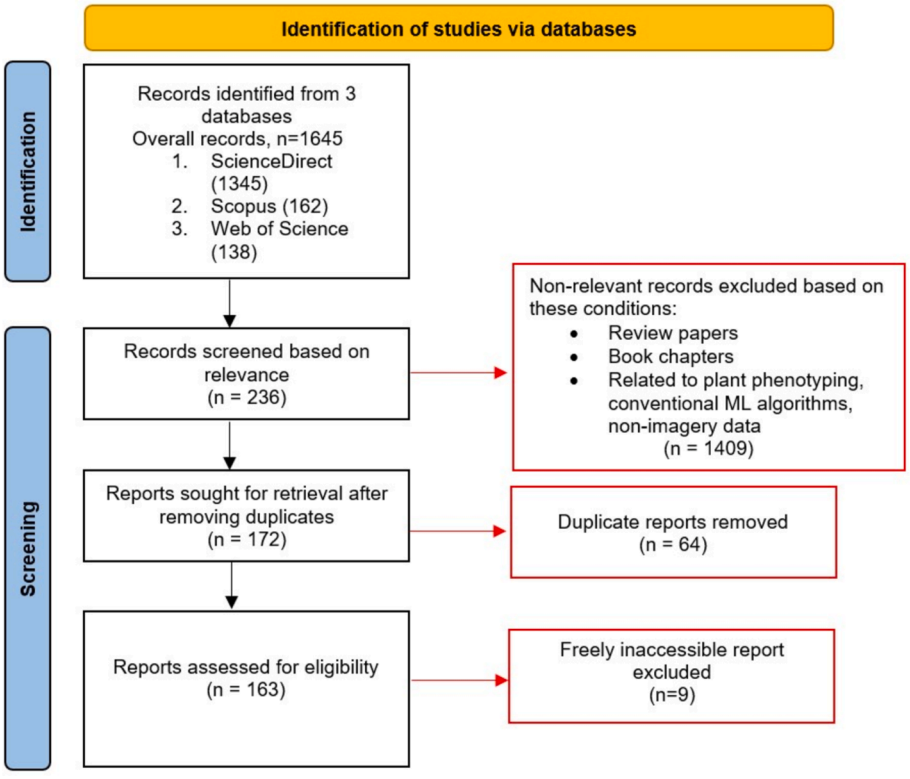


Fig. 2. Systematic literature selection using PRISMA guidelines (Haddaway et al., 2022).

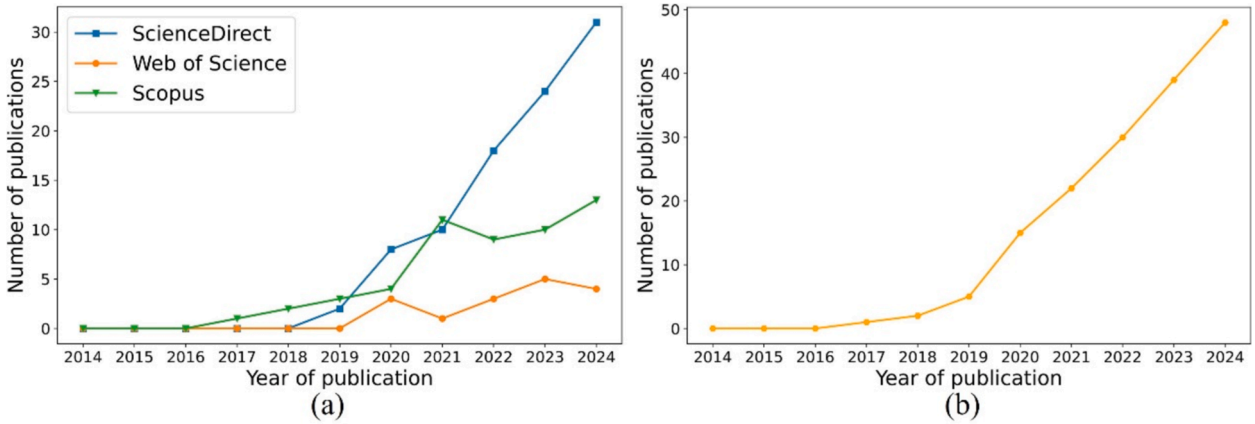


Fig. 3. Trend of publications in the area of deep learning and image-based plant stress detection (a) in each database (b) Overall.

Additionally, it demonstrates that the overall trend of publications is rising with each passing year (Fig. 3b).

3.2. General deep learning and image-based plant stress detection system

3.2.1. Image acquisition

Image acquisition for plant stress detection involves capturing high-quality images of plants using various imaging techniques and devices. Choosing appropriate imaging equipment based on the specific requirements of application is critical for building a successful computer vision-based stress detection system. Adjustment of imaging parameters or conditions such as exposure time, aperture, focal length, resolution, lighting condition, camera position, and image background is important to ensure high quality training data for the system.

In recent years, most commonly used imaging techniques for plant stress detection are RGB cameras (Anami et al., 2020; Zhuang et al., 2020), multispectral cameras (Blekanov et al., 2023; Zhou et al., 2021c), hyperspectral cameras (Khotimah et al., 2023; Vincke et al., 2023), and thermal cameras (Bompilwar et al., 2022; S. Khan et al., 2019). RGB sensors, operating on the principles of Red, Green, Blue, are extensively utilized for plant stress determination due to their capacity to capture changes in color and light intensity reflected by plants. However, their capability to capture spectral data beyond the visible range is limited (Tang et al., 2023). Conversely, multispectral cameras can capture images across numerous discrete spectral bands (Elmasry et al., 2019), enabling the detection of subtle variations in plant health and stress that may not be discernible to the human eye. Similarly, hyperspectral cameras, which capture images across hundreds or thousands of contiguous spectral bands, possess the capability to analyze entire spectral signatures. As a result, it can discriminate between different

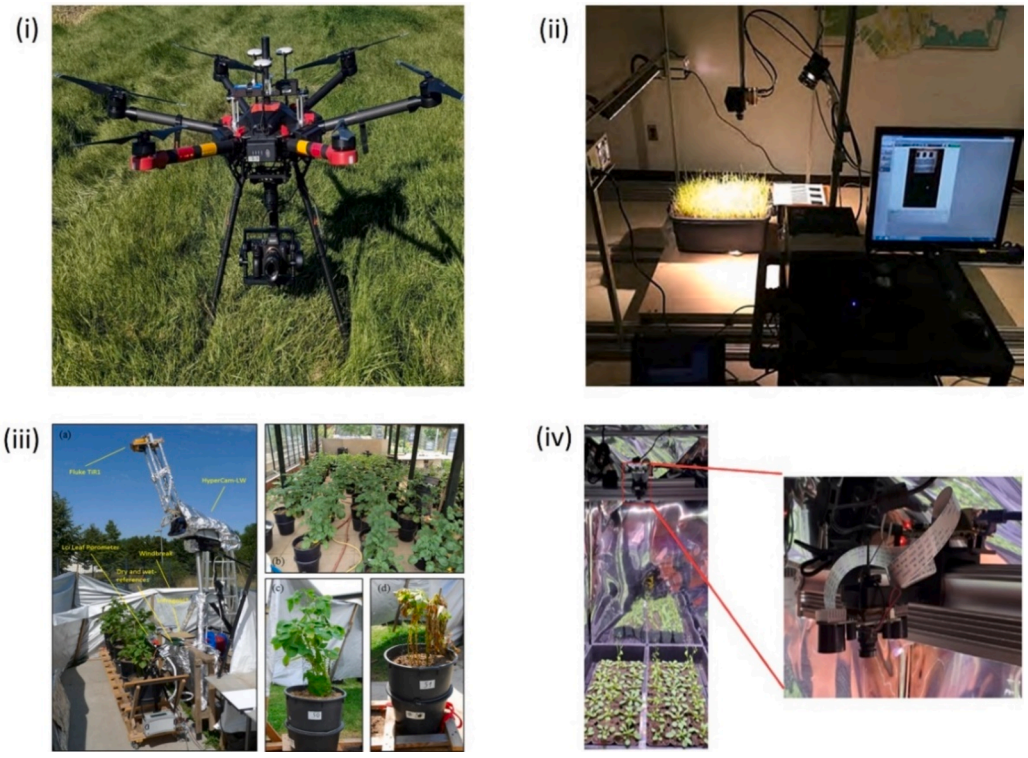


Fig. 4. Image acquisition setups using different sensors (i) DJI Matrice 600 Pro with a Sony Alpha 7R II, 42.4-megapixel RGB camera mounted on it(Sapkota, 2021), (ii) A close-range laboratory imaging system with a Micro-Hyperspec VNIR sensor in controlled lighting condition (Dao et al., 2021a), (iii) HyperCam on the tripod, Fluke TiR1, Lci leaf porometer, Infragold as well as dry and wet references targets (Gerhards et al., 2016) (iv) Chamber equipped with two Raspberry Pi 3B + and an ArduCam Noir Camera with a motorized IR-CUT filter and two infrared LEDs (Sakeef et al., 2023).

materials and identify subtle spectral signatures associated with specific stressors or physiological changes in plants (Sarić et al., 2022). Furthermore, thermal cameras, which detect and capture images based on thermal radiation emitted by objects, provide valuable insights into temperature differentials, heat distribution, and thermal anomalies caused by both biotic and abiotic stressors (Wen et al., 2023). Additionally, other data acquisition devices such as smartphones (Esgario et al., 2020a; Zhao et al., 2022a), hand-held cameras (Hu et al., 2023), Raspberry Pi cameras (Chandel et al., 2024; Sakeef et al., 2023), and laser displacement sensors (Wakamori et al., 2020) have been utilized for plant stress detection purposes.

Unmanned Aerial Vehicle (UAV) (Blekanov et al., 2023; J. Li et al., 2020; Zhou et al., 2021c) platforms equipped with cameras have also emerged as effective tools for collecting large-scale field data (Fig. 4i). Sujatha et al., (2023) utilized UGVs equipped with night vision cameras to monitor the agricultural lands. Blekanov et al. (2023) used the Geoscan-401 UAS and Wagimin et al., (2024) used UAV equipped with the Micasense RedEdge multispectral cameras for aerial photography. Sankararao et al., (2021) used a Resonon's Pika-L HSI camera mounted on a DJI matrice-600 Pro UAV for water stress monitoring in Chickpea. Turkulainen et al., (2023) used a custom-built quadcopter UAS with a Gryphon Dynamics frame equipped with dual-RGB camera, Rikola hyperspectral camera, and MicaSense RedEdge to monitor bark beetle-induced spruce damage. Choudhury et al., (2022) utilized a DJI Mavic Mini 2 lightweight drone as the UAV to gather overhead images of a tea garden and Debnath & Saha, (2022) took images of diseased rice paddy using DJI tello drone camera. Zwieback et al., (2024) and Virnodkar et al., (2021) used satellite images of Maxar Worldview-2 and Sentinel-2, respectively as their data for DL model evaluation. Moreover, several studies utilized publicly available plant image datasets, including, PlantVillage (Anim-Ayeko et al., 2023a), Plant Village Paddy, DiaMOS dataset (Thokala and Doraikannan, 2023), and PlantCV (Azimi et al., 2021a). Fig. 5 shows that most researchers have used publicly available sources for their data and among the imaging sensors (RGB, multispectral, hyperspectral, and thermal), RGB was the most used data acquisition method compared to other imaging technologies so far.

3.2.2. Data preprocessing steps

Preprocessing of image data is an essential step in preparing it for deep learning models. In general, the plant images are acquired from the open field. As a result, it might contain several artifacts and noise, which need to be eliminated before deep learning application (Munjal et al., 2023). Additionally, proper preprocessing techniques can help improve the model's training speed, performance, and generalization ability. Common preprocessing techniques involve image resizing, normalization/ standardization, data augmentation, and noise removal.

Image resizing is essential for standardizing input sizes, which improves computational efficiency, performance (Saponara and Elhanashi, 2022) and ensures compatibility with the model architecture. Nearest neighbor interpolation (Cándido-Mireles et al., 2023) is the most basic interpolation method for image resizing that interpolates the nearest pixel to the current pixel. It takes substantially shorter processing time than other methods because this method simply looks for the nearest neighbor and computes one pixel at a time. Another way of resizing images is manually cropping the image (Anami et al., 2020; Zhuang et al., 2020), where the region of interest such as a specific feature of the plant is isolated and irrelevant background information is removed. For example, Bompilwar et al. (2022) center-cropped the images to remove the text present in the images and Grunig et al., (2021) center-cropped to isolate leaf from background. Shewale & Daruwala, (2023) performed corner crop, where they reduced the resolution of the central region to 512 × 512 to preserve the most detailed background. Then, the image was divided into four 96 × 96-pixel chunks. However, in some cases, center, corner, and random crop are not appropriate as it may lose important information such as leaf wilting area. Wakamori et al. (2020) cropped the images based on optical flow that detects leaf wilting and crops it from the original image. They determined the cropped area to maximize the daily average optical flow magnitude, which was calculated using DeepFlow. Nayak et al., (2023) executed image resizing by resampling using the pixel-area relationship.

In the context of image data, the normalization or standardization of pixel values typically involves the scaling of these values to a standardized range or distribution, commonly encompassing the interval [0,

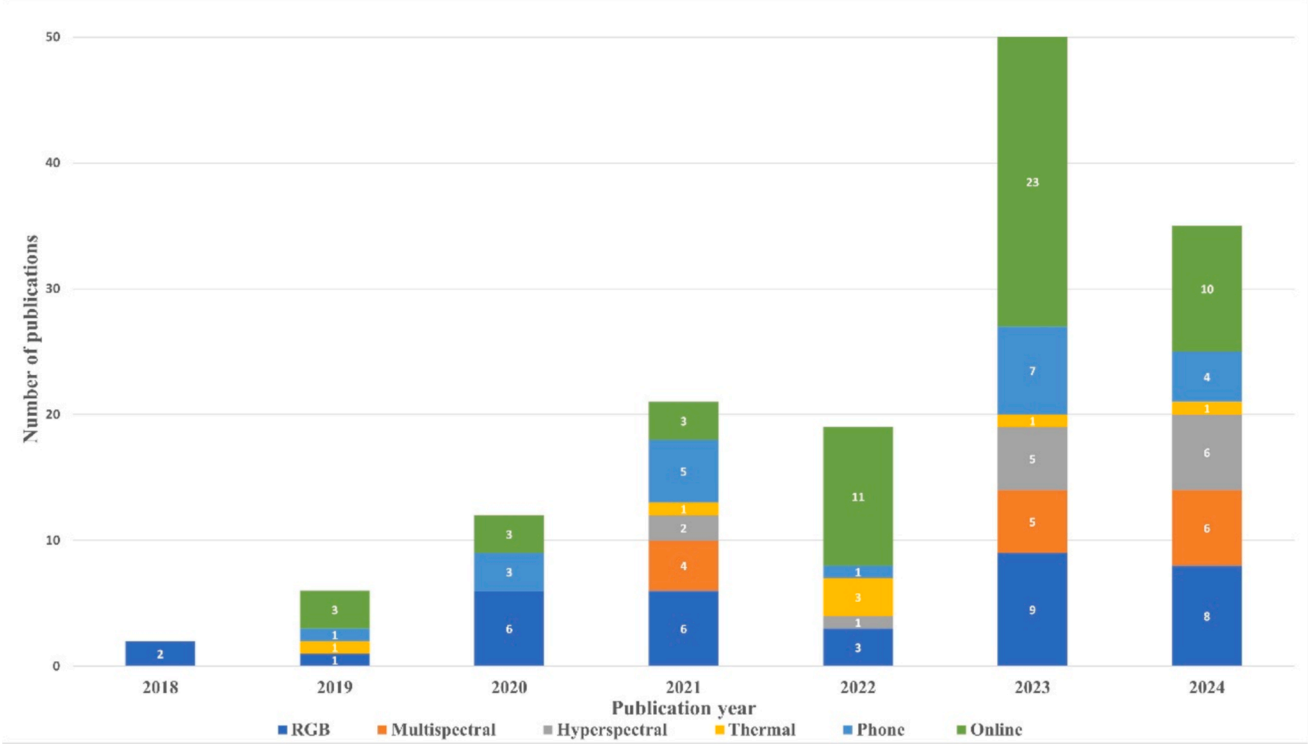


Fig. 5. A bar chart showing the number of articles using each image acquisition systems/ sources each year.

1] or [-1, 1]. Normalization helps transform input data into the ranges for the sigmoid activation functions that allows faster gradient descent convergence (Dao et al., 2021a) and this step helps reduce the computing costs for deep learning model (Thokala and Doraikannan, 2023). Most common techniques include dividing all image pixels by the value 255 to convert the pixel values to the 0 to 1 range (Anim-Ayeko et al., 2023a; N et al., 2021), min–max normalization (Dao et al., 2021a; Sakeef et al., 2023), and rescaling the data to maintain a mean of 0 and a standard deviation of 1 (Khotimah et al., 2022). Blekanov et al. (2023) used "Calculator-Grid normalization" tool to normalize the 5 layers (red, green, blue, near infrared, red edge) of the images.

Image augmentation is another important preprocessing step to artificially increase the size of the dataset by generating new, realistic-looking images from the existing ones. Fig. 6 visualizes some common augmentation techniques, such as, randomly cropping an image, increasing or decreasing brightness, applying a blurring filter that smooths image details by averaging pixel intensities (gaussian blur), adding random pixel noise to the image (gaussian noise), altering the contrast of the image by making dark regions darker and bright regions brighter, or vice versa, shearing along the X axis by shifting image pixels in the X-direction by an amount proportional to their Y-coordinate, and flipping the image along horizontal or vertical axis. In addition to mitigating data scarcity, it captures variability in plant appearance and addresses class imbalance (Bompilwar et al., 2022). Some standard data augmentation methods previously used by researchers are image rotation, flipping, and shifting (Asaari et al., 2023; Thokala & Doraikannan, 2023). Furthermore, zooming (Franchetti and Pirri, 2022; Thokala and Doraikannan, 2023), gaussian blur (Franchetti and Pirri, 2022), shear transform, skewing, random noise (Anim-Ayeko et al., 2023a), color variation (brightness, contrast, and saturation) (Esgario et al., 2020b), PCA jittering (Karlekar & Seal, 2020), resizing (Tassis & Krohling, 2022), salt and pepper noise, noise rejection (Chamara et al., 2023), luminosity (Ji & Wu, 2022), and image sharpening (Hao et al., 2020) have also been used in recent studies. When analyzing the contrast value in colour augmentation, Reddy et al., (2023) used histogram equalization to provide a more consistent result. Khotimah et al. (2022) used the Synthetic Minority Oversampling technique (SMOTE) to oversample the minority class to overcome the class imbalance problem. Moreover, Esgario et al. (2020b) explored the use of a new approach proposed by Zhang (2017), which combines two randomly selected images to generate a new one. Nazki et al., (2020) proposed AR-GAN to address data imbalance, which is the extended GAN-based image-to-image translation method by enhancing the perceptual quality of the generated images.

Noise removal is another important preprocessing step for image data analysis. RGB format is somewhat noisier and image intensity separation is difficult. Additionally, these images make it difficult to extract image features. Hence, Nandhini et al., (2022) and Lu et al., (2023) converted the images into HSV color space that describes colors in terms of Hue, Saturation, and Value. The HSV color model is often preferred over the RGB model where color description plays an integral role. Additionally, Sathyavani et al., (2022) adopted median filter to eliminate noise present in 3 × 3 windows of RGB images and Raj et al., (2022) employed convoluted Gaussian filtering to remove noise.

3.2.3. *Deep learning (DL) techniques*

DL has shown to be highly successful in plant stress detection due to its ability to learn complex patterns, automatically extract informative features from images or spectral data and availability of large-scale datasets. A general DL model works through a series of layers that process data, learn patterns, and make predictions based on those patterns. Typically, a DL architecture consists of input layer, hidden layers (convolutional layers, recurrent layers, or fully connected), and an output layer as shown in Fig. 7. The model begins with an input layer that receives preprocessed data. The hidden layers, consisting of neurons, learns features of the data by applying a series of weights and biases to the input. Activation functions (such as ReLU, sigmoid, or softmax) introduce non-linearity, enabling the network to learn complex patterns in data. Additionally, the error minimization is performed through backpropagation and optimization. Finally, the output layer provides the model's prediction.

3.2.3.1. DL models suitable for small training data. Transfer learning (TL) is a technique in ML in which knowledge learned from a task is reused in order to boost performance on a related task (Gauridhumal, 2023). TL has become popular in plant stress classification due to its ability to leverage pretrained models to improve performance, provide faster training, and address data scarcity. As shown in Table 1, several studies have utilized pre-trained CNN models for feature extraction where the model is initialized using the weights learned from another large dataset (Azimi et al., 2021; Franchetti and Pirri, 2022; Li et al., 2018; Rojanarungruengporn and Pumrin, 2021). N et al., (2021) used ImageNet database to pre-train InceptionResNetV2 model for rice leaf diseases prediction and Dawod & Dobre, (2021) used Cifar10 dataset to pre-train ResNet-152 model for classifying sunflower foliar diseases. COCO dataset was used by Kanerva et al., (2022) to pre-train YOLOv4-p5

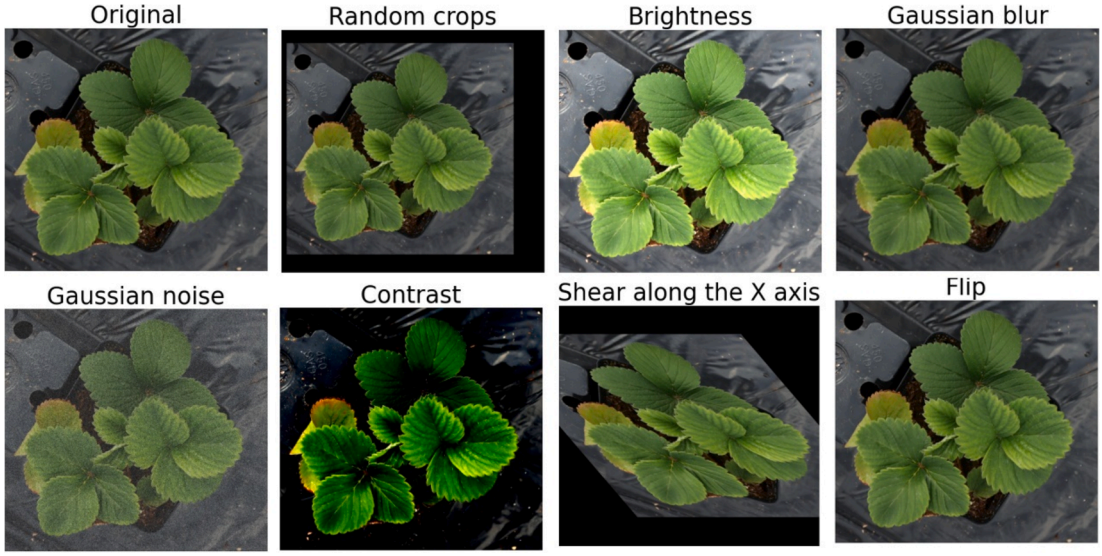


Fig. 6. Examples of data augmentation techniques.

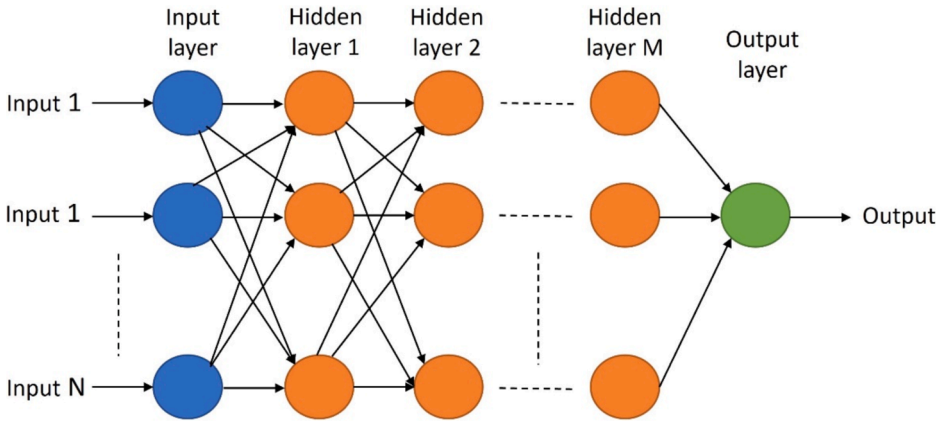


Fig. 7. A general deep learning (DL) architecture.

model and by Loyani, (2024) to pre-train Mask RCNN model. Malik et al., (2024) proposed a novel CNN model, TreeNet and pre-trained the model on PlantVillage dataset.

Usually during TL model training, the weights of the pretrained models are frozen and the last layer is replaced to match the number of classes in the target task. H. Li et al. (2018) truncated the last layer of the CNN model, which is a softmax layer that targets at 1,000 classes of the ImageNet dataset, and replaced it with a new softmax layer that targets at two classes (the drought condition and the control condition). Azimi et al., (2021) replaced the last layer with LSTM classifier to predict water stress category using the features extracted from Inception-v3 and VGG16. Abisha & Bharathi, (2023) used pre-trained Xception model for feature extraction and random forest (RF) for classification. Rojanarungruengporn and Pumrin (2021) used LSTM cell with multi-layer perceptron (MLP) to classify extracted features to determine the stress of the plants. Selection of model architecture for feature extraction can vary with the input data and task, for example, H. Li et al. (2018), Anami et al. (2020), and Rojanarungruengporn and Pumrin (2021) used VGG-16, whereas Azimi et al., (2021) used Inception-v3 along with VGG-16 and Franchetti and Pirri, (2022), Malvade et al., (2022), Zwieback et al., (2024) and Grunig et al., (2021) used ResNet-50 architecture for feature extraction where these models were pre-trained on a large dataset.

Rangarajan Aravind & Raja, (2020) evaluated six pre-trained CNNs namely AlexNet, VGG16, VGG19, GoogLeNet, ResNet101 and DenseNet201 and demonstrated GoogLeNet having the best validation accuracy of 97.3 %. Fenu & Malloci, (2022) compared pre-trained MobileNetV2, EfficientNetB0, InceptionV2, VGG16 and ResNet50 models and MobileNetV2 showed the best performance. Dey et al., (2022) compared pre-trained VGG-19, VGG-16, ResNet50, and InceptionV3 models to detect biotic and abiotic diseases in rice plants and VGG-19 showed the highest prediction accuracy of 91.8 %. Simhadri & Kondaveeti, (2023) evaluated 15 pre-trained CNN models, DarkNet-53, DenseNet-201, GoogLeNet, Inceptionv3, MobileNetv2, ResNet-50, ResNet-101, ShuffleNet, SqueezeNet, Xception, InceptionResNetV2, NasNetMobile, VGG16, AlexNet, and EfficientNetB0 for the automatic identification of rice leave diseases. Their results show that the InceptionV3 model was the most effective with an average accuracy of 99.64 %. Chandel et al. (2024) utilized TL-based model GoogLeNet, which outperformed AlexNet, Inception-v3, MobileNetV2, and ResNet-50 for successfully identifying water stress in plants. Esgario et al. (2022) used several pretrained CNN architectures, such as AlexNet, GoogLeNet, VGG16 and ResNet-50 and showed that ResNet-50 had the best result with an accuracy of 97.07 %. Asaari et al. (2023) used the pre-trained YOLOv3 (darknet53.conv.74), pre-trained YOLOv4 (yolov4.conv.137) and the pre-trained SSD MobileNet V2 FPNLite 320 × 320 model, where they froze all layers, except for the last three high-level layers for fine-tuning. Li et al., (2020) utilized unsupervised pre-training on a large set of unlabeled data acquired by UAS to reduce the issue of limited labeled images and enhance the robustness of the soybean IDC rating model. First, they pre-trained a convolutional autoencoder (CAE) on unlabeled sub-images clipped from aerial RGB images to initialize the CNN model. Then, this pretrained CNN model was trained again on the labeled plot-wise sub-images clipped from stitched RGB maps. Their results indicated that the unsupervised pre-training of CAE on the unlabeled data alleviated the required number of the labeled training data.

Few-shot learning (FSL) is a ML framework in which an AI model learns to make accurate predictions by training on a very small number of labeled examples. It is typically used to train models for classification tasks when training data is scarce. Tassis and Krohling (2022) evaluated few shot learning methods by using Prototypical Networks and Triplet Networks for coffee leaves biotic stress classification and severity estimation. They reported that FSL had better performance (93.25 % accuracy) than the state-of-the art methods for severity estimation task. Self-supervised learning (SSL) is another paradigm of ML model, which does not depend on external labels provided by humans. It is suitable when there is limitation of labeled data. SSL is trained on a task using the data itself and leverage inherent structures or relationships within the input data to generate supervisory signals. Nagasubramanian et al. (2022) compared seven SSL approaches in plant stress detection and reported that pretraining on unlabeled plant stress images significantly outperformed TL methods using random initialization for soybean stress classification. They also observed that SSL methods were robust to class imbalances, highly efficient and able to differentiate confounding stress classes with a smaller number of training images compared to supervised learning methods with ImageNet initialization. Wang et al., (2023) proposed STNet, which was pre-trained in a SSL framework, that is, a huge amount of unlabeled pixel-level spectral data were used to pre-train STNet's backbone network to improve the generalization of extracted features. Later, the labeled average spectral data were used to fine-tune STNet to identify tomato BW severity.

3.2.3.2. State-of—the-art DL models. Several CNN architectures have gained popularity for plant stress detection tasks (Table 2) due to their effectiveness in capturing hierarchical features from plant images. Some examples are, DenseNet-BC (Sathyavani et al., 2022), ResNet (Gómez-Zamanillo et al., 2023; Picon et al., 2019b; Anim-Ayeko et al., 2023a), LeNet (Albanese et al., 2021; Cruz et al., 2017), U-Net (Giakoumoglou et al., 2024; J. Zhang et al., 2020), and YOLO (Javierto et al., 2021; Shahriar Zaman Abid et al., 2024). However, the choice of the ideal DL architecture for a specific task depends on the data and task itself. One of the ways to find the ideal model is to evaluate multiple models and compare their performances. Cándido-Mireles et al. (2023) presented an assessment of five CNN architectures, VGG16, ResNet-50, Inception-v3, EfficientNetB3, and LeNet, for detecting plant stress in grapevine RGB

Table 1
Summary of articles using TL-based deep learning models.

Serial no.	Article	Data acquisition	Plant species	Stress type	Model	Result
1	H. Li et al., 2018	RGB camera (not specified)	Maize, sorghum	Drought	CNN, VGG-16, BRNN, LSTM, BLSTM	BLSTM was best model with 80.3 % precision, 77.3 % recall and 78.8 % F-score
2	Rançon et al., 2019	RGB 2592 × 2048 camera	Vineyards	Fungal disease esca	MobileNet	91 % accuracy
3	Rangarajan Aravind and Raja, 2020	Smartphone	Eggplant, hyacinth beans, ladies finger and lime	Ten different diseases	AlexNet, Visual Geometry Group 16 (VGG16), VGG19, GoogLeNet, ResNet101 and DenseNet201	GoogLeNet had best validation accuracy of 97.3 %
4	Anami et al., 2020	Nikon D3300 Digital SLR camera (RGB)	Paddy crop	Bacterial blight, Fungal blast, Drought, Submergence, Nitrogen deficit, Phosphorus deficit, Potassium deficit, Boron deficit, Zink deficit, Iron deficit, Chemical injury	VGG-16 CNN	Average classification accuracy 92.89 %
5	Li et al., 2020	UAS consisted of a six-rotor Matrice 600 Pro platform and an onboard Zenmuse X5R RGB camera	Soybean leaf	Iron Deficiency Chlorosis (IDC)	CAE1-CNN, CAE2-CNN, Ran-CNN	CAE-CNN (R^2 ranged from 0.48 to 0.70) exhibited higher robustness than the baseline Ran-CNN model
6	Mishra et al., 2020	Smartphone	Corn	Rust and Northern Leaf blight	DCNN	Average accuracy of 98.40 %.
7	Dawod & Dobre, 2021	iPhones	Sunflower	Rust, downy mildew, powdery mildew, alternaria leaf blight	ResNet152	Average accuracy of Rust: 0.929, downy mildew:0.529, powdery mildew:0.717, alternaria leaf blight:1
7	Grunig et al., 2021	Smartphone	Apple orchards	Mines of blister moth, Physical damages, Brown spots, Lepidoptera, Mildew, Feeding	ResNet50	Accuracy of 93.8 %
8	N et al., 2021	Publicly available	Rice leaf	Leaf blast, brown spot, and bacterial blight	InceptionResNetV2	Accuracy of 95.67 %
9	R. Sujatha et al., 2021	DSLR (RGB)	Citrus leaf	Blackspot, Melanose, Canker, greening	Inception-v3, VGG-16, VGG-19	VGG-16 had best accuracy of 89.5 %
10	Azimi et al., 2021	Canon EOS 60D (RGB)	Chickpea plant shoot	Water stress	CNN-LSTM	Accuracy of 98.52 % on JG-62 and 97.78 % on Pusa-372 chickpea plant data
11	Rojanarungruengporn and Pumrin, 2021	Publicly available	Sorghum	Nitrogen deficiency	VGG16_LSTM	85 % accuracy
12	Malvade et al., 2022	Nikon D3300 Digital SLR camera (RGB) and publicly available	Paddy	Brown Spot, Hispa pests, Leaf Blast	Inception-V3, VGG-16, ResNet-50, DenseNet-121 and MobileNet-28	ResNet-50 had highest accuracy of 92.16 %
13	Gautam et al., 2022	Publicly available	Paddy	Blight, blast, brown spot	InceptionV3, VGG16, ResNet, SqueezeNet, and VGG19	96.43 % accuracy
14	Kanerva et al., 2022	Samsung NX1000, Samsung NX300, Sony A7R, Sony A7R II, UAS, Trimble Geo XT GPS-device	Norway spruce	European spruce bark beetle	YOLOv4-p5	F-score 89 %
15	Fenu & Malloci, 2022	Publicly available	Multiple species	Multiple stresses	EfficientB0, MobileNetV2, InceptionV2, ResNet50 and VGG16	MobileNetV2 had best test accuracy on Laboratory dataset which is 99.31 % and on field dataset which is 87.02 %
16	Dey et al., 2022	12MP autofocus camera sensor (not specified)	Rice	Hispa pest, two fungal diseases, and NPK deficiency	5 layers CNN, InceptionV3, ResNet50, VGG16, VGG19	Accuracy of 91.8 %
17	de Melo et al., 2022	Thermal FLIR® camera fitted to a cellphone	Sugarcane	Water deficits	Inception-Resnet-v2	0.889 F-score
18	Esgario et al., 2022	Not specified	Arabica coffee leaves	Leaf miner, rust, brown leaf spot and cercospora leaf spot	AlexNet, GoogLeNet, VGG16 and ResNet-50	ResNet-50 had best result with accuracy of 97.07 %
19	Amogi et al., 2023	Thermal infrared (Lepton 3.5, Teledyne FLIR LLC), RGB (OV5647, OmniVision technologies), weather sensors (ATMOS14 and	Apple	Heat stress	Mask R-CNN	91.4 % average precision

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Table 1 (*continued*)

Serial no.	Article	Data acquisition	Plant species	Stress type	Model	Result
		ATMOS41, METER Group)				
20	Chamara et al., 2023	Meidas SL122 trail cameras, OV5642 imaging sensors with ArduCAM camera shields, Raspberry Pi Camera Module V2 with Raspberry Pi Zero	Corn and soybean	Insect	Custom DCNN	Mean average precision of 0.02
21	Kiruba & Arjunan, 2023	CAT S62 Pro smartphone	Paddy	Bacterial Leaf Blight (BLB), Bacterial Leaf Streak (BLS), Bacterial Panicle Blight (BPB), Black Stem Borer (BSB), Blast, Brown spot, Downy Mildew, Hispa, Leaf Roller, Tungro, White Stem Borer, Yellow Stem Borer	VGG16, MobileNet, Xception, and ResNet34	ResNet34 had highest F1-score of 97.50 %
20	Simhadri & Kondaveeti, 2023	Publicly available	Rice	Bacterial leaf blight, Brown spot, Hispa, Leaf blast, Leaf scald, Leaf streak, Narrow brown spot, Sheath blight, Tungro	DarkNet-53, DenseNet-201, GoogLeNet, Inceptionv3, MobileNetv2, ResNet-50, ResNet-101, ShuffleNet, SqueezeNet, Xception, InceptionResNetV2, NasNetMobile, VGG16, AlexNet, and EfficientNetB0	InceptionV3 had best accuracy 0f 99.64 %
21	Singh et al., 2023	Not mentioned	Rice	Bacterial leaf blight, Brown spot, and Leaf smut	MobilNet	Accuracy rate of 87 %
22	Vijayarangan et al., 2023	Publicly available	Rice	Brown Spot, Bacterial Blight, and Leaf Blast	Inception V3	Accuracy of 98.74 %
23	Abisha & Bharathi, 2023	Not mentioned	Banana leaf	Multiple stress types	VGG 16, RESNET, MobileNet, InceptionResNetV2, Inception, and Xception	Xception-RF had 100 % accuracy
24	Kang et al., 2023	iPhone 8 Plus	Cotton	Verticillium wilt disease	DarkNet-19, VGG-16, ResNet-50, and InceptionV3	VGG-16 had best accuracy of 0.98
25	Nayak et al., 2023	iPhone X, Xiaomi Redmi Note 8	Rice	12 rice diseases and deficiency	DenseNet201, Xception, MobileNetV2, and ResNet50	DenseNet201 had best accuracy of 0.9803
26	Abbasi et al., 2023	Publicly available	Lettuce, Basil, Parsley, Spinach	Bacterial leaf spot and Downy mildew, Septoria leaf spot, Stemphylium leaf spot	ResNet-50 and YOLOv5s	95.83 % accuracy and 82.13 mAP
27	Wang et al., 2023	Hyperspectral (Specim IQ camera)	Tomato	Wilt	Transformer-based network (STNet)	91.93 % accuracy
28	K. Sujatha et al., 2023	UGVs with night vision cameras (RGB)	Paddy	False smut (FS), Sheath blight (SB), Rice blast (RB), Leaf scald (LS), Brown spot (BS), Bacterial leaf blight (BLB) and Bakane (BE)	AlexNet, DenseNet, ResNet	AlexNet had best accuracy of 99.16 %
29	Hu et al., 2023	Canon EOS 80D hand-held camera (RGB)	Tea leaf	Leaf blight	SVM, AdaBoost, VGGNet, ResNet, GBM	GBM had best F1-score of 0.77
30	Asaari et al., 2023	Not specified	Eggplant plant	Diverse stress symptoms	YOLOv3, YOLOv4, SSD	YOLOv4 was best with mean Average Precision of 0.83
31	Chandel et al., 2024	Canon PowerShot SX740, Raspberry Pi camera, and three different smartphone cameras (Apple iPhone 11, MI Note 4, and Samsung Galaxy M21)	Wheat and maize crops	Water stress	AlexNet, GoogLeNet, Inception V3, MobileNet V2, and ResNet 50	GoogLeNet had best Accuracy of wheat (92.9 %) and maize (97.9 %)
32	O'Halloran et al., 2024	Not mentioned	Maize	Necrosis, Streak Virus disease	EfficientNet V2 B0 and B1, VGG16, ResNet50, and LeNet-5	EfficientNet V2 B0 had best accuracy of 99.99 %
33	Zwieback et al., 2024	Satellite (Maxar Worldview-2), multispectral	Spruce	Spruce beetle	DeepLabV3, ResNet-50	Accuracy of 95 %
34	Mazumder et al., 2024	Publicly available	Banana leaf, black gram leaf	Segatoka, Xamthomonas, Anthracnose, Leaf Crinckle, Powdery Mildew, Yellow Mosaic	DenseNet201Plus	0.9012 accuracy
35	Wagimin et al., 2024	UAV equipped with a MicaSense RedEdge 3 multispectral camera	Mango	Chlorophyll	InceptionV3, DenseNet121 and ResNet50	InceptionV3 had best accuracy of 96.49 %
36	Loyani, 2024	Canon EOS Kiss X7 and Samsung SM-G570F cameras	Tomato	Tuta absoluta's damage	Mask RCNN	mAP of 85.67 %

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Table 1 (*continued*)

Serial no.	Article	Data acquisition	Plant species	Stress type	Model	Result
37	I. Khan et al., 2024	Publicly available	Maize	9 diseases	VGGNET, InceptionV3, ResNet50, InceptionResNetV2	ResNet50 had best accuracy of 87.51 %
38	Rezaei et al., 2024	CANON EOS 2000D DSLR camera (RGB)	Barley	Net form net blotch (NFNB), spot form net blotch (SFNB), and scald	DenseNet, ResNet, InceptionV3, Xception, and MobileNet	MobileNet had best accuracy of 98.63 %
39	Kapetas et al., 2024	Multi-spectral (customized Qcell Phenocheck camera)	Tomato	Grey mould disease	YOLOv8	81.7 % Mean Average Precision (mAP)
40	Malik et al., 2024	Publicly available	Corn	Multiple diseases	TreeNet	99.8 % accuracy
41	Supreetha et al., 2024	Publicly available	Rice	NPK deficiency	InceptionV3, VGG16, VGG19, ResNet50, and ResNet152	Accuracy ranges from 91 % to 99 %
42	Bouskour et al., 2024	Canon EOS (RGB)	Wheat Leaf	Stripe Rust and Septoria	MobileNetV2	Accuracy of 0.99
43	Orka et al., 2024	iPhone 13 Pro Max	Ash gourd, bitter gourd, and snake gourd	Cucurbits nutritional deficiency	EfficientNetB0, MobileNetV2, and DenseNet121	90 % test accuracy with DenseNet121

images and their results indicated that EfficientNetB3 exhibited the best result with a 97.2 % accuracy. Sakeef et al. (2023) found that ConvLSTM2D performed comparatively better than other DL algorithms, such as FCN, ResNet, LSTM, Encoder, and Conv2D. Blekanov et al. (2023) used five state-of-the-art neural network architectures based on U-Net models (U-Net, Attention U-Net, R2-UNet, Attention R2-Unet, and U-Net3+) to solve the nitrogen level rate segmentation issue for crops on an orthophotomap. Their experiment showed that Attention R2U-Net was more robust and reliable among all models for different kinds of crops with accuracy of 97.59–99.96 %. Shin et al., (2021b) evaluated AlexNet, SqueezeNet, GoogLeNet, ResNet-50, SqueezeNet-MOD1, and SqueezeNet-MOD2. Their results indicated that ResNet-50 performed best among all models with a classification accuracy of 98.11%. Giakoumoglou et al., (2023) compared AlexNet, DenseNet-121, EfficientNet B0 and B1, GoogLeNet, MobileNet v3 small, ResNet-18, ResNet-34 and ResNet-50, VGG-11 and VGG-11 with Batch Normalization (BN) and showed that VGG-11 (BN) provided the best accuracy of 0.932. Christakakis et al., (2024) used state-of-the-art U-Net++ architecture for feature extraction and to boost the extraction abilities, it was paired with MobileViTv2-1.25 encoder, which improved model efficiency.

3.2.3.3. Custom DL models. Several studies designed custom DL architectures for plant stress detection, as summarized in Table 3. These custom designs suit the specific characteristics of the task and the available data. These architectures proposed by the studies have proved to outperform the state-of-the-art DL architectures in terms of efficiency, overfitting and accuracy. Azimi et al., (2021a) proposed a 23-layered DL technique (Fig. 8v) and compared it with traditional ML techniques and DL architectures. Their results revealed that this architecture is comparable to the established state-of-the-art deep learning architectures like ResNet-18 and NasNet Large. Islam and Yamane, (2021) introduced HortNet417v1 with 417 layers for rapid recognition, classification, and visualization of plant stress conditions in real time and their model proved to have high accuracy and a low computing requirement (Fig. 8ii). A comparison with some state-of-the-art models indicated that the proposed model performed better than Xception, ShuffleNet, and MobileNetv2 in terms of overfitting. Bompilwar et al. (2022) proposed a novel CNN based architecture with 5 CNN layers, 2 max-pooling layers, and fully connected or dense layers, which outperformed three state-of-the-art pre-trained models, namely VGG-16, ResNet-50, and MobileNetV2.

DCNN- based models.

Deep Convolution Neural Network (DCNN)- based models are widely utilized in image-based stress detection tasks due to their ability to effectively capture and interpret spatial relationships within images. A unique DCNN was chosen by Ghosal et al. (2018) after an exhaustive exploration of various applicable DCNN architectures and their classification and explanation capabilities (Fig. 8i). This proposed architecture consists of 5 convolutional layers (128 feature maps of size 3*3 for each layer), pooling layers (down-sampling by 2*2 max-pooling), and 4 batch normalization layers and 2 fully connected (FC) layers with 500 and 100 hidden units each, sequentially. They observed that their proposed model captured color-based features at the low abstraction levels (i.e., first or second convolution layer). This conforms to the general observation of deep neural networks capturing simple low-complexity but important explainable features at the lower layers (Balu et al., 2017; Shrikumar et al., 2017). Ghosal et al. (2018) also noticed that the learnt model seems to be agnostic to species, which demonstrates the ability of TL. A double-staged deep CNN (D^2CNN) was proposed by Swaminathan & Vairavasundaram, (2024), which consists of two independent expert-level classification models that work together. In the first stage, the subsequent stages will not be processed as soon as the classifier predicts the input leaf as healthy. Otherwise, the model will pass to the second stage, where the customized CNN with deep layers classifies the input image into one of the ten paddy leaf stress categories. Kaur et al., (2024a) proposed Hybrid Deep Segmentation Convolutional Neural Network Model (Hybrid-DSCNN) for semantic segmentation of leaf disease that is the combination of the Convo layer of U-Net and SegNet models. There are thirty-six convo layers used in the U-SegNet model which were changed in the Hybrid-DSCNN model by replacing 7 Conv layers of U-Net and 13 Conv layers of SegNet models in down-sampling. They also used the pooling indices (features' vector) from the encoder feature and transfer them for mapping to the corresponding up-sampling layer.

RCNN- based models.

The region-based convolutional neural network (RCNN) extends traditional CNNs by incorporating region proposals, enabling object detection and localization tasks. RCNN uses the selective search method to generate several candidate regions in an image and the Fast RCNN is a more sophisticated form of RCNN, which uses a multi-task loss function for performing classification and regression tasks based on CNN. Zhao et al. (2022b) proposed a specified Faster RCNN algorithm for stress detection of tea plants at a canopy scale. A CNN was used to extract the features from each candidate region generated by the selective search method. Then, these features were used as the input of SVM for category determination and a linear regression model was used for position refinement. The authors also assessed the impact of using multiple combinations of image bands (such as RGB, NearIR, RedEdge, and NDVI). The results revealed that the combination of RGB, NearIR, and NDVI channels had the best performance and the use of the RedEdge band showed a significant negative impact on the segmentation quality of the nitrogen level in the agricultural field. Mehzabeen & Gayathri, (2025) designed an adaptive-based Mask Region-Based Convolutional

Table 2
Summary of articles using state-of-the-art deep learning models.

Serial no.	Article	Data acquisition	Plant species	Stress type	Model	Result
1	Cruz et al., 2017	Mobile device or camera (not specified)	Olive tree	Olive Quick DeclineSyndrome (OQDS)	LeNet	98.60 % ± 1.47 accuracy
2	Picon et al., 2019a	Not mentioned	Wheat	Rust, Septoria, and Tan Spot	Modified ResNet-50	Average BAC of 0.98
3	Picon et al., 2019b	Cell phone	Multi-crop (wheat, barley, corn, rice and rape-seed)	Septoria tritici, Puccinia striiformis, Puccinia recondita, Septoria nodorum, Drechslera tritici-repentis, Gibberella zeae, Blumeria graminis and Oculimacula yallundae, Helminthosporium turcicum, Phoma lingam, Pyrenophora teres,Ramularia collo-cygni, Rhynchosporium secalis, Puccinia hordei, Thanatephorus cucumeris, Pyricularia oryzae and dirty panicle	ResNet-50	For single crop, average BAC of 0.92, for multi-crop, average BAC of 0.93
4	Kavitha Lakshmi & Nickolas, 2020	Not mentioned	Betelvine	Not specified	Mask RCNN	Mean average precision of 84.07 %
5	J. Zhang et al., 2020	Matrice 600 UAV, a Nikon D800 (RGB)	rapeseed	Nitrogen Stress	U-Net	F-measure of 91.56 % and IoU of 83.19 %
6	Nagasubramanian et al., 2020	Not mentioned	Soybean	8 stress types	DenseNet-121	95.05 % accuracy
7	Albanese et al., 2021	RaspberryPi microcomputer equipped with a camera, an Intel Neural Compute Stick (NCS)	Not mentioned	Pest	LeNet-5, VGG16, MobileNetV2	VGG16 had best accuracy of 97.9 %
8	Sankararao et al., 2021	Resonon's Pika-L HSI camera, DJI matrice-600 Pro UAV	Chickpea	Water stress	3D-2D CNN	Accuracy of 95.44 %
9	Javierto et al., 2021	Smartphones	Robusta Coffee	Not specified	YOLOv3	90 % accuracy
10	Shin et al., 2021b	DSLR camera: EOS 1300D (RGB)	Strawberry	Powdery mildew disease	AlexNet, SqueezeNet, GoogLeNet, ResNet-50, SqueezeNet-MOD1, and SqueezeNet-MOD2	ResNet-50 gave the highest CA of 98.11 %
11	Tassis et al., 2021	Publicly available	Coffee leaves	Leaf miner, leaf rust, and cercospora leaf spot	Mask R-CNN	71.90 % average precision
12	Dao et al., 2021a	Micro-Hyperspec VNIR sensor	Bromus inermis	Drought stress	Deep learning Multilayer Perceptron	94.6 % – 100.0 % accuracy
13	Choudhury et al., 2022	DJI Mavic Mini 2 lightweight drone as UAV	Tea	Red Spider Mites	YOLOv5	IOU of 99.7 %
14	Ji & Wu, 2022	Publicly available	Grape	Black measles	ResNet50-based DeepLabV3+	Accuracy of 97.75 %
15	Kumi et al., 2022	Not mentioned	Cocoa	Black Pod and Swollen Shoot	CenterNet ResNet50 V2, EfficientDet D0, SSD MobileNet V2, and SSD ResNet50 V1 FPN	SSDResnet50 V1 FPN performed best with a mean average precision (mAP) of 32.4 %
16	Sathyavani et al., 2022	Kaggle repository	Rice crop	Nutrient deficiency	DenseNet-BC	Test accuracy 96 %
17	Turkulainen et al., 2023	RGB (Sony A7R and Sony A7R II), multispectral (Micasense Altum, Micasense RedEdge), Hyperspectral (Rikola camera), Trimble's APX-15 EI UAV GNSS-Inertial OEM positioning system	Spruce	Bark Beetle-Induced Spruce Damage	2D-3D-CNN	F1-score of 0.759
18	Ruan et al., 2023	Hyperspectral (Surface Optics Corporation (SOC)-701-SWIR)	Tomato	Freeze and drought stress	ProtoNet, RelationNet	Accuracy of 69.57 %
19	M. Li et al., 2023	Mobile Phone	Rapeseed	Nutrient deficiency	PSPNet, DeepLab v3+, U-Net, Attention block	UNet-Attention had best accuracy of 90 %
20	Giakoumoglou et al., 2023	Multispectral (Qcell MUSES9-MS camera)	Cucumber	Grey mould	AlexNet, DenseNet-121, EfficientNet B0 and B1, GoogLeNet, MobileNet v3 small, ResNet-18, ResNet-34 and ResNet-50, VGG-11 and VGG-11 with Batch Normalization (BN)	VGG-11 (BN) had best accuracy of 0.932
21	Blekanov et al., 2023	Geoscan-401 UAS, DJI Matrice 600pro, Pika L hyperspectral camera, Micasense RedEdge MX multispectral camera	Crops (Not specific)	Nitrogen nutrition	U-Net, Attention U-Net, R2U-Net, Attention R2U-Net and U-Net3 +	Attention R2U-Net was more robust and reliable for different kinds of crops (accuracy 97.59–99.96 %)

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Table 2 (*continued*)

Serial no.	Article	Data acquisition	Plant species	Stress type	Model	Result
22	Gómez-Zamanillo et al., 2023	High-resolution camera (5184 × 3456 pixels) (not specified)	Soybean and redroot amaranth plants	Herbicide damages	ResNet-34	R^2 of 0.87 for soybean and 0.89 for redroot amaranth
23	Anim-Ayeko et al., 2023a	Publicly available	Potato, tomato	Blight disease	ResNet-9	Test accuracy of 99.25 %
24	Cándido-Mireles et al., 2023	Smartphones (Xiaomi MI CC9, Huawei ALE-L23, and iPhone 8), Parrot Sequoia cameras	Grapevine	Not specified	EfficientNetB3	97.2 % accuracy
25	Sakeef et al., 2023	ArduCam Noir Camera	Arabidopsis thaliana	Light stress	FCN, Conv2D, ConvLSTM2D, Encoder, LSTM, ResNet	ConvLSTM2D had best accuracy of 80 %
26	Giakoumoglou et al., 2024	Multispectral (Qcell1 Phenocheck camera)	Cucumber	Botrytis cinerea symptoms	U-Net++	Intersection over Union (IoU) of 0.656
27	Shahriar Zaman Abid et al., 2024	Publicly available	Rice, corn, wheat, potato, and tomato	10 stress types	YOLOv8	mAP of 98 %
28	Guo et al., 2024	Publicly available	Not mentioned	40 pest classes	ResNet8	Accuracy of 84.29 + 0.23 %
29	Kim et al., 2024	Raspberry Pi version 3, Pi RGB camera module 2	Bentgrass	Drought	Predefined CNN in Tensorflow-keras version 2.13	Accuracy of 0.90 to 0.95
30	Rezaei et al., 2024	Publicly available	14 crop species	Multiple stress types	ViT and ResNet50	> 90 % accuracy
31	Bao et al., 2024	Imec Snapscan VNIR hyperspectral camera	Sugarcane	Smut and mosaic diseases	ResNet34	90 % accuracy
32	Saleem et al., 2024	Publicly available	Soybean	Eight different soybean stresses	ResNet50	95.09 % accuracy
33	Rajwade et al., 2024	Thermal (Krykard TCA 1950), RGB (Canon EOS 3000D)	Okra	Water Stress	ResNet-50 and MobileNetV2	ResNet-50 had best accuracy of 87.9 %
34	Christakakis et al., 2024	Multispectral (customized Qcell Phenocheck camera)	Cucumber	Botrytis cinerea	Combination of U-Net++ with MobileViTv2-1.25	Mean Dice Similarity Coefficient (mDSC) of 0.792
35	Yao et al., 2024	Not mentioned	Winter wheat	Drought stress	DenseNet121	Accuracy of 94.67 %
36	Rajpal et al., 2024	Publicly available	11 crop species	Multiple stress types	Inception-ResNet-V2-based model	Accuracy of 0.9991 ± 0.002
37	Wang et al., 2024b	Intel RealSense D435i RGB-D (Red-Green- Blue-Depth) camera, RedEdge-MX multispectral camera	Poplar seedling	Drought stress	ResNet18-LSTM and ResNet18-CBAM-LSTM	99.69 % accuracy
38	Upadhyay & Gupta, 2024	Publicly available	Apple, guava, and custard apple crops	Fungal diseases	Modified ResNeXt CNN model	98.92 % accuracy

Neural Network (MRCNN) for rice disease classification. It contains backbone network, Region Proposal Network (RPN), mask representation and ROI align. MRCNN returns the image with labeled data, coordinates with a bounding box of disease-affected regions in images, and provides the spatial layout in terms of masking the spot in leaf images.

GAN- based models.

Generative Adversarial Networks (GANs) are highly effective in computer vision because they generate realistic synthetic data and enhance various computer vision tasks. A GAN is a DL model consisting of two neural networks, called the generator (G) and the discriminator (D), that compete against each other in a game-theoretic setup. The G takes a random gaussian noise as input and produces fake data as output, whereas the D takes in either real or generated data as input and produces a corresponding probability score as output. Mahadevan et al., (2024) proposed deep spectral GAN (DSGAN^2) for enhancing data in detecting rice plant leaf disease. The generator was trained to generate new rice leaf images that closely resembles the raw images. The algorithm's first step was identifying the healthy and non-healthy leaves. Then, the gain weight of disease and non-disease plant images was estimated. The next step was to find their actual weight to produce output to categorize the leaves as diseased or non-diseased. Zhu et al., (2024) proposed cDCGAN to generate labeled HSI samples for data augmentation, which follows the principle of cGAN while adopting the architecture of DCGAN. DCGAN is a variant of GAN that introduces convolution and deconvolution layers in the D and G, respectively, without spatial pooling, upsampling and fully connected hidden layers. cGAN is a conditional version of GAN, by introducing an additional input to condition on both the G and D, which enables the simultaneous generation of pseudo data with designated class labels. cDCGAN is capable of generating high-quality data with labels simultaneously. Nazki et al., (2020) introduced AR-GAN, where in addition to the adversarial loss, the synthetic image generator optimizes on Activation Reconstruction loss (ARL) function that optimizes feature activations against the natural image. Their results showed a significant improvement in classification accuracy (+5.2 %) using generated synthetic samples as compared to (+0.8 %) increase using classic augmentation in an equal class distribution environment.

CAM- based models.

Channel Attention Modules (CAMs) are primarily used to enhance features within CNNs by selectively weighting feature channels. Khotimah et al. (2022) proposed a novel DL method, named Spectral Convolution and Channel Attention Network (SC-CAN) (Fig. 8iv). It comprises two main modules, a spectral convolution module and a channel attention module. The spectral convolution module consists of dilated causal convolutional layers stacked in a residual manner in order to capture the spectral features. These layers can learn both local and global features and also have long receptive fields, which makes them suitable for capturing long dependency patterns in hyperspectral data. The channel attention module contains a global pooling layer and fully connected layers that compute the inter-relationship between the feature map channels before scaling them based on their importance level (attention score). This module weights the feature maps according to their importance level since not every feature map generated by the dilated convolutional layers is significant. The results showed that SC-

Table 3
Summary of articles using custom build deep learning models.

Serial no.	Article	Data acquisition	Plant species	Stress type	Model	Result
1	Ghosal et al., 2018	Canon EOS Rebel T5i, 18 mega-pixels (RGB)	Soybean	Bacterial blight, Septoria brown spot bacterial pustule, SDS, frog-eye leaf spot, herbicide injury, potassium deficiency, and IDC	DCNN	94.13 % accuracy on unseen test dataset
2	Venal et al., 2019	Publicly available (Github resources)	Soybean	biotic (e.g., fungal and bacterial diseases), abiotic (e.g., nutrient deficiency and chemical injury)	Hybrid CNN-SVM	97.11 % test accuracy
3	S. Khan et al., 2019	Seek thermal compact camera	Tomato	Water and light stress	Supervised neural network	Not mentioned
4	Karlekar & Seal, 2020	Publicly available database PDDB	Soybean	Anthracnose, Bacterial blight, Carijo Leaf, Charcoal Rot, Copper Phytotoxicity, Leaf Cercospora, Mela/ Rhizoctonia aerial blight, Mildio/Downy mildew, Murcha Sclerocio, Oidio/ Powdery mildew, Phytophora rot, Rust, Septoria/Brown Spot, Southern blight	SoyNet	98.14 % accuracy on test dataset
5	Nazki et al., 2020	Publicly available	Tomato	Nutritional excess or deficiency, Powdery mildew, Gray mold, Plague, Canker, Whitefly, Leaf mold, Low temperature, Miner	AR-GAN	47.15 FID
6	Hao et al., 2020	Sony ILCE-5000L/A5000 mirrorless camera (RGB)	Purple leaf lettuce	Light stress	MFC-CNN	87.95 % accuracy
7	Esgario et al., 2020b	Smartphones (ASUS Zenfone 2, Xiaomi Redmi 5A, Xiaomi S2, Galaxy S8, and iPhone 6S)	Coffee leaf	Biotic stresses: leaf miner, rust, brown leaf spot, and cercospora leaf spot	Modified ResNet-50	95.24 % accuracy for the stress classification and 86.51 % accuracy for severity estimation
8	Zhuang et al., 2020	CCD digital camera with a type of Panasonic WV-SW396AH (RGB)	Maize	Water stress	CNN-SVM	88.41 % AWTR and 93.87 % AWSR
9	Wakamori et al., 2020	RGB cameras (GoPro HERO5 Session, GoPro Inc.), wireless scattered light sensor nodes, and laser displacement sensors (HL-T1010A, Panasonic Corporation)	Tomato leaf	Water stress	Multimodal neural network with C-Drop	0.429 R^2
10	Venal et al., 2020	Publicly available (Github resources)	Soybean leaf	Biotic: fungal and bacterial abiotic: nutrient deficiency and chemical injury	CNN-Softmax, CNN-SVM	CNN-Softmax had 98.04 % accuracy and CNN-SVM had 99.59 % accuracy on test data
11	Zhou et al., 2021b	multispectral camera RedEdge-M, GPS, infrared (IR) thermal camera, Raspberry Pi 3B, UAV (DJI Matrice 600 Pro)	Soybean	flooding injury	Custom FNN	90 % accuracy
12	Virnodkar et al., 2021	Sentinel-2 virnoite image from ESA's Copernicus website	Sugarcane	Water stress	DenseResUNet	61.91 % mIoU, and 80.53 % accuracy
13	Su et al., 2021	DJI Matrice 100, RedEdge camera	Wheat	Yellow Rust	Custom U-Net	91.3 % precision
14	Butte et al., 2021	UAV Solo by 3DR, multispectral camera Sequoia by Parrot	Potato crop	Not mentioned	Retina-UNet-Ag	Average dice score coefficient (DSC) of 0.74
15	J. Gao et al., 2021	RGB camera (Sony RX 100 iii)	Potato	Late blight lesion	Custom SegNet	IoU values of background (soil and leaf) and lesion classes in the test dataset are 0.996 and 0.386, respectively
16	Sharma et al., 2021	Not mentioned	Rice, potato leaf	Blight	Custom CNN	99.58 % accuracy for rice data, 97.66 % accuracy for potato leaves data
17	Noguera et al., 2021	DJI™ Matrice 100 UAV, MicaSense RedEdge-M™ multispectral camera, GPS device, downwelling light sensor (DLS)	Olive	Nitrogen, Phosphorus, and Potassium (NPK)	Custom ANN	R^2 of 0.63 for N, 0.89 for P, and 0.93 for K

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Table 3 (*continued*)

Serial no.	Article	Data acquisition	Plant species	Stress type	Model	Result
18	Zhou et al., 2021a	Not mentioned	Tomato and cucumber	Tomato powdery mildew, tomato early blight, cucumber powdery mildew, cucumber virus 10disease, cu11cumber downy mildew, and cucumber bacterial spot	ITK-Net	99.63 % accuracy
19	Zhou et al., 2021c	UAV platform (DJI Matrice 600 Pro), RGB camera (GoPro Hero 5 Black), multispectral camera (PowerShot ELPH 300, Canon & MaxMax.com-LDP LLC), CHDK	Soybean	Drought stress	Mixed CNN	R^2 of 0.78
20	Azimi et al., 2021a	Canon EOS 60D (RGB)	Sorghum	Nitrogen deficiency	Custom 23 −layered CNN	84 % accuracy
21	Islam and Yamane, 2021	Smart phone (iPhone 11)	Peach trees	Water stress	HortNet417v1	93 % test accuracy
22	Tejasri et al., 2022	UAV platform DJI Inspire-1 Pro, Zenmuse X5 RGB sensor	Maize	Drought stress	Ensemble U-Net and U-Net++	Mean IoU of 0.71
23	Joshi et al., 2022	Smartphone (Samsung Galaxy M11) and digital camera (Nikon D5600 DSLR)	Rice	Bacterial and fungal	RiceBioS	Test accuracy of 93.25 %
24	Russel & Selvaraj, 2022	Publicly available	Multiple species	Multiple stresses	Custom CNN	98.61 % accuracy
25	Vasantha et al., 2022	Publicly available	Rice	8 disease classes	RDD_CNN	98.47 % test accuracy
26	Guo-feng et al., 2022	Publicly available	Strawberry	15 diseases	LFC-Net	92.48 % accuracy
27	Kaur et al., 2022	Publicly available	Tomato leaf	Bacterial_spot, Leaf_mold, Septoria_spot, Target_spot, Mosaic_virus, Early_blight.	Mask R-CNN	Mean Average Precision (mAP), F1-score, and accuracy of 0.88, 0.912, and 0.98, respectively
28	Raj et al., 2022	Not mentioned	Tomato and brinjal leaves	Cercospora Leaf Spot and Bacterial Blight, Powdery Mildew and Rust	Deep convoluted active contour neural network (DCACNN)	98 % accuracy
29	Belay et al., 2022	Techno 7 mobile camera	Chickpea	Ascochyta blight and Fusarium wilt	CNN-LSTM	Accuracy of 92.55 %
30	Wongchai et al., 2022	Publicly available	Multiple species	Multiple stress types	DACL RNN	Accuracy of 96 %
31	Pandey & Jain, 2022	Publicly available	44 plant types	Multiple diseases	Deep attention dense CNN (DADCNN)	Accuracy of 99.93 % and 97.33 % for two databases
32	Debnath & Saha, 2022	DJI tello drone camera	Rice paddy	Brown spot disease	Custom CNN	Accuracy of 97.701 %
33	Feng et al., 2022	Visible near-infrared HSI system consisted of imaging spectrograph and CCD camera, tungsten halogen lamps, conveyer belt	Cabbage	Clubroot	CNN1 and CNN2	Accuracy of 85 %.
34	Nandhini et al., 2022	Not mentioned	Plantain tree	Xanthomonas wilt (BXW), Bunchy top disease (BBTD), Black sigatoka (BS), Yellow sigatoka (YS)	Gated-Recurrent Convolutional Neural Network (G-RecConNN)	Accuracy of 0.960 ± 0.006
35	Zhao et al., 2022b	MI 6 smartphone, iPhone XR smartphone camera	Tea Plant	Green leafhopper, anthracnose and leaf sunburn	Faster RCNN	Mean average precision (mAP) was 76.07 %
36	Khotimah et al., 2022	Publicly available	Chinese Spring (CS), Aegilops columnaris (co(CS)), Ae. speltoides auchery (sp(CS)), 15Kharchia and Fusarium	Salt stress, Fusarium head blight disease	Spectral Convolution and Channel Attention Network (SC-CAN)	Overall accuracy of 83.08 %, 88.90 %, 82.44 %, 82.10 %, and 82.78 % on CS, co(CS), sp(CS), Kharchia, and Fusari
37	Franchetti and Pirri, 2022	HR digital camera (RGB)	Lettuce	Tip-burn stress	Weakly-supervised semantic segmentation methods (WSSS)	meanIoU 0.74 and F1 0.85
38	Bompilwar et al., 2022	FLIR E8 thermal camera (Kaggle)	Not specific	Bacteria Leaf Blight, Blast, Leaf Spot, Leaf Folder, Hispa	tCrop Lite	94 % test accuracy
39	Zeng et al., 2023	RGB (SONY DSC-RX100M6 digital camera), DJI Mavic 2 Pro UAV	Sweet potato	Sweet potato virus disease (SPVD)	SPVDet, SPVDet-Nano	Accuracy of 78.1 %
40	Jha & Shukla, 2023	Nikon D5300 camera (RGB)	12 diverse tree leaves	Not specific	DCNN	Accuracy of 96.38 %
41	Latif et al., 2023	Publicly available	Coffee Leaf	Cercospora, Phoma, Rust	ECNN, HLGGM, HLGCM	HLGGM had best accuracy of 99.49 %
42	Nampally et al., 2023	Trimble R10 GNSS Receiver, DJI Inspire-1 Pro UAV, Micasense RedEdge-MX multispectral camera,	Maize	Water stress	StressNet	Validation accuracy of 91.30 %

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Table 3 (*continued*)

Serial no.	Article	Data acquisition	Plant species	Stress type	Model	Result
		Downwelling Light Sensor (DLS)				
43	Salamai & Al-Nami, 2023	Smartphones	Coffee leaf	Leaf miner, brown leaf spot, rust, and Cercospora leaf spot	Mixed vision transformer (MVT) learner	Accuracy of 98.51 % ± 2.09 for disease classification and 96.13 % ± 2.17 for severity estimation
44	Lu et al., 2023	iPhone12, Redmi K30Pro, and Huawei P30Pro	Rice	Rice blast, Sheath blight, Brownspot, and Leaf blight	CNN-BiGRU	Accuracy of 98.21 %
45	Mao et al., 2023	UAV (DJ M200 V2), multispectral camera (MS600) and thermal imager (Zenmuse XT2)	Tea	Cold stress	CNN-GRU	Rp^2 of 0.862
46	Daniya & Vigneshwari, 2023	Publicly available	Rice	BLB, Blast, and Brown spot	RWW-based NN	Accuracy of 0.893
47	Divyanth et al., 2023	Not mentioned	Corn	Gray Leaf Spot (GLS), Northern Leaf Blight (NLB), and Northern Leaf Spot (NLS)	Integration of UNet and DeepLabv3+	R^2 value of 0.96
48	Reddy et al., 2023	Publicly available	Rice, potato and tomato	Multiple diseases	Customized PDICNet	99.73 % accuracy
49	Ahmad et al., 2023	Publicly available	Corn and potato	Multiple diseases	Branched neural network architecture (BNNDC)	99.40 % accuracy
50	Shaodan et al., 2023	UAV (DJI Mavic 2 Pro), RGB (L1D-20c camera)	Rice	Blast	RiceblastSegMask	97.56 % accuracy
51	Pudumalar & Muthuramalingam, 2023	Publicly available	Cotton	Areolate, Mildew, Sore Shin, Fusarium, Wilt, and Myrothecium	Hydra	Accuracy of 95 %
52	Shewale & Daruwala, 2023	Samsung A52S mobile camera and publicly available	Tomato	9 biotic and abiotic stresses caused by fungi, bacteria, virus and infections	Custom CNN	0.9981 accuracy
53	Milke et al., 2023	Sony Cyber-shot DSC-W800 digital camera (RGB)	Coffee	Wilt	Custom CNN	97.9 % test accuracy
54	Khotimah et al., 2023	Hyperspectral sensor (PIKA II, Resonon)	Wheat and cassava	Salt stress, cassava brown streak and mosaic virus disease	MCE-ST	Average accuracy 85.29 ± 0.26 on salt stress data and 98.64 ± 1.60 on cassava data.
55	Thokala & Doraikannan, 2023	Publicly available	14 species	17 fungal diseases, diseases caused by bacteria, mold, virus and mites	DCNN-MCViT	Accuracy of 99.82 %
56	Lin et al., 2024	Huawei Honor v20 mobile phone	Peanut	leaf rust disease and leaf spot disease	Improved YOLOv8n	89.80 % mAP
57	Swaminathan & Vairavasundaram, 2024	Publicly available	Paddy	15 classes of biotic and abiotic stresses	D^2CNN	Accuracy of 95 %
58	Macdonald et al., 2024	Publicly available	Multiple species	Multiple stress types	Custom CNN	96.09 % validation accuracy
59	Xiao et al., 2024	Hyperspectral imager (SOC710VP)	Maize	Nicosulfuron toxicity	Multi-branch deep learning model	91.37 % accuracy
60	Subeesh & Chauhan, 2024	Smartphonecamera (Xiaomi, sony IMX)	Tomato	Leaf miner pests	Attention-based CNN	Accuracy of 97.87 %
61	Vallabhajosyula et al., 2024	Publicly available	Multiple species	Multiple stress types	Custom model using Improved Vision Transfomer and ResNet9	99.7 % accuracy
62	Bezabh et al., 2024	Not mentioned	Mango	Leaf and fruit rot diseases	GoogLeNet, and VGG16 ensemble CNN	99.21 % accuracy
63	J. Zhang et al., 2024	Hyperspectral (UHD185 imaging spectrometer)	Rice	Bacterial blight disease	ISMSFuse	98.14 % accuracy
64	Liu et al., 2024	Publicly available	Maize	Common rust, gray spot, blight	MResNet	97.45 % accuracy
65	Wang et al., 2024a	RGB (BM-500GE color camera)	Apple	Alternaria Boltch, Brown Spot, Mosaic and Rust	Deep evidence fusion framework	98.1 % accuracy
66	Tejaswini et al., 2024	Publicly available	Tomato, potato, and bell pepper leaf	14 stress types	Custom CNN	Accuracy of 86.21 %
67	X. Zhou et al., 2024	Hyperspectral (sCMOS camera)	Oilseed rape	Lead content	Transfer stacked convolution auto-encoder (T-SCAE)	Rp^2 of 0.9374
68	Zhu et al., 2024	SNAPSCAN VNIR (visible and near-infrared) hyperspectral camera	Strawberry	Drought stress	cDCGAN	0.91 accuracy
69	Mahadevan et al., 2024	Publicly available	Rice	Nine disease categories	$DSGAN^2$	Accuracy of 97 %,

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Table 3 (*continued*)

Serial no.	Article	Data acquisition	Plant species	Stress type	Model	Result
70	Jesie et al., 2024	Canon Power Shot G2 digital camera (RGB)	Paddy	Bacterial Leaf Blight (BLB), Sheath Rot (SR), Narrow Brown Leaf Spot (NBLS), Brown Spot (BS) and Leaf Smut (LS)	HCNN	97 % accuracy
71	Abisha & Bharathi, 2024	Not mentioned	Banana, tomato, and bean leaves	Not specific	Modified Xception-RF	Above 99 % accuracy
72	Venkatesh et al., 2024	Publicly available, Raspberry Pi equipped with a mounted camera module	Groundnut plant leaves	Nitrogen, phosphorus, and potassium stress	Hybrid quantum–classical model (HQCM)	97.79 % accuracy
73	Kaur et al., 2024b	Publicly available	Tomato leaves	Moisture, insects, temperature, nutrients, fungus, bacteria, and water influence disease patterns	Hybrid Deep Segmentation Convolutional Neural Network Model (Hybrid-DSCNN)	Accuracy of 98.25 %
74	Pal et al., 2024	Publicly available	Rice	BrownSpot, Leaf Blast, Hispa	Custom CNN	Accuracy of 97.9 %
75	Bera et al., 2024	Publicly available	Banana, Coffee, Potato, Apple, Tomato, Potato, Strawberry, Soybean, Raspberry, Grapes, Corn, Bell-pepper, and others	Nutrition deficiency and multiple diseases	Plant Nutrition Deficiency and Disease Network (PND-Net)	(a) 90.00 % Banana and 90.54 % Coffee nutrition deficiency; and (b) 96.18 % Potato diseases and 84.30 % on PlantDoc datasets
76	Goyal et al., 2024	Sony DSC-HX400V RGB camera	Maize	Drought stress	Custom-CNN	98.53 % accuracy on test dataset
77	Kaur et al., 2024a	Publicly available	Tomato	Moisture, insects, temperature, nutrients, fungus, bacteria, and water influence ailment forms on the leaves	hybrid Deep Segmentation Convolutional Neural Network (Hybrid-DSCNN)	98.24 % accuracy

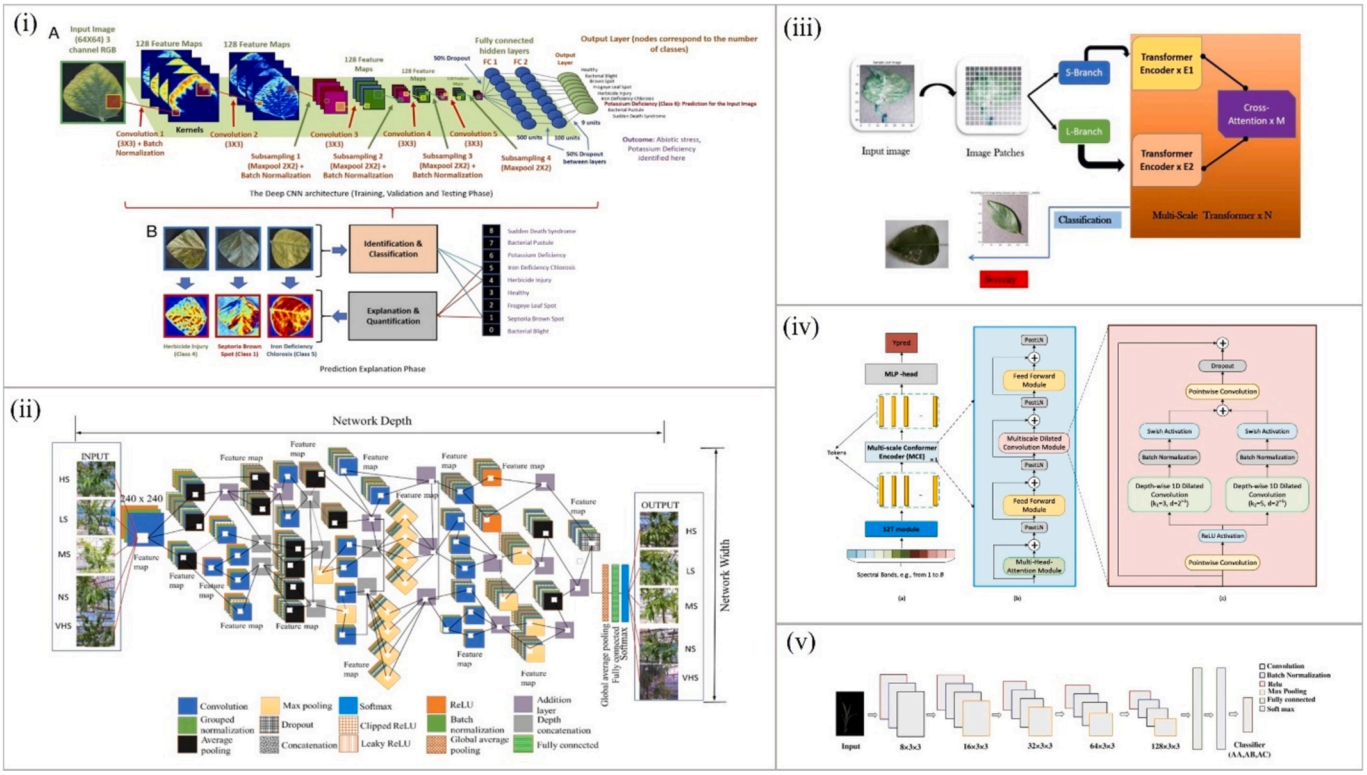


Fig. 8. Examples of custom CNN architectures (i) Overall schematic of the xPlNet framework: (A) DCNN architecture (B) Explanation phase (Ghosal et al., 2018), (ii) HortNet417v1 network architecture (HS-high stress, LS-low stress, MS-moderate stress, NS-no stress, VHS—veryhigh stress) (Islam and Yamane, 2021), (iii) Proposed DCNN with CrossViT (DCNN-MCViT) (Thokala & Doraikannan, 2023), (iv) (A) Overview of the proposed network comprising three main blocks, (B) details of the Multiscale Conformer Encoder block, and (C) details of the multiscale convolution module (Khotimah et al., 2022), (v) Proposed custom architecture (Azimi et al., 2021a).

CAN outperformed the existing state-of-the-art methods, 1D CNN, LSTM, spectralFormer, and sRN. Subeesh & Chauhan, (2024) proposed a novel approach that uses both spatial and channel attention modules to enhance the feature extraction capability of the CNN, which outperformed the state-of-the-art CNN models AlexNet, VGG16, and VGG19 due to its feature extraction abilities. Pandey & Jain, (2022) introduced a new deep attention dense convolutional neural network (DADCNN) for plant leaf disease classification. In this study, the attention learning weights are determined from the trained network without adding an extra learnable layer in the CNN module for attention learning. As the higher layer features possess more ability for semantic separation than the lower layer features, the output features of the higher layer were used as attention mask of the lower layer.

Transformer- based models.

Transformer models rely on self-attention mechanisms to capture relationships across all parts of a sequence, making them highly effective for computer vision tasks. Unlike traditional models like RNNs, which process data sequentially, transformers process all elements in a sequence simultaneously, allowing them to capture long-range dependencies with ease and achieve parallelization. Khotimah et al. (2023) proposed MCE-ST, a convolution-transformer (conformer) based framework to classify hyperspectral image data. This model was capable of exploiting the complementary strengths of transformers and CNNs as they are able to capture both short- and long-range band dependencies of hyperspectral images. It used a convolution-based tokenization method to extract local dependency between spectral bands in contrast to the conventional transformer, which uses a linear projection for tokenization. Moreover, since different hyperspectral samples may have different spans of local relationships, a multiscale conformer encoder (MCE) comprising two separate branches of depth-wise dilated convolution with different kernel sizes was used to extract the different spans of the local interactions between tokens. Thokala and Doraikannan, (2023) developed DCNN-MCViT, which leverages the emerging capabilities of Vision Transformers (VT) (Fig. 8iii). This model employed a multi-scale vision transformer with cross-attention for efficient detection and classification of plant stress. A DCNN was constructed and ViT was added to it, which was presented as CrossViT (DCNN-MCViT). Salamai & Al-Nami, (2023) presented a mixed vision transformer (MVT) learner to generate mixed contextual attention maps from discriminatory latent representations between support and query images to give more emphasis to the biotic stress lesions in coffee leaves. Vallabhajosyula et al., (2024) proposed a model for early detection of plant diseases using a novel hierarchical residual vision transformer with improved Vision Transformer and ResNet9 models. The model can extract more meaningful and discriminating features by reducing the number of trainable parameters with a smaller number of computations. The results showed that the proposed model was able to outperform other state-of-the-art models such as InceptionV3, MobileNetV2, and ResNet50.

GRU- based models.

Gated Recurrent Unit (GRU) models are a type of RNN models designed to have simple structure and overcome some limitations of traditional RNNs, particularly vanishing and exploding gradients. GRUs use gating units to control the flow of information through the network, which helps them manage long-term dependencies and avoid vanishing gradients. Lu et al., (2023) proposed CNN-BiGRU that combines CNN and RNN for diagnosing rice diseases. They employed an improved CNN module as an image feature encoder and BiGRU to learn relevant information from both forward and backward directions. Mao et al., (2023) improved a hybrid CNN-GRU network model, which retains the feature extraction advantages of CNN and the time series data mining advantages of GRU. They stated that the CNN-GRU hybrid models outperformed the single DL model (GRU).

U-Net- based models.

U-net architecture has been the backbone of custom DL models of several studies (Butte et al., 2021; Divyanth et al., 2023; Su et al., 2021; Tejasri et al., 2022; Virnodkar et al., 2021). Its encoder-decoder structure, along with skip connections, enables it to capture both local and global features effectively. Butte et al., (2021) proposed a novel DL model for detecting crop stress, named Retina-UNet-Ag, which is a variant of Retina-UNet and includes connections from low-level semantic representation maps to the feature pyramid network. Retina-UNet integrates features that are characteristical for U-Net into the RetinaNet model for object objection. Divyanth et al., (2023) integrated UNet and DeepLabv3 + to accurately estimate the severity of targeted diseases and resulted in a R^2 value of 0.96. Virnodkar et al., (2021) proposed a novel architecture 'DenseResUNet' for identifying water-stressed sugarcane crops using segmentation based on encoder-decoder approach. The novelty of the proposed approach lies in the replacement of classical convolution operation in the UNet with the dense block. The traditional convolutional layers were replaced with the DenseRes block that utilizes the power of deep residual networks and densely connected networks. Secondly, the reduction in features maps is performed using a down transition block instead of max pooling in UNet and restructured the decoder path with DenseRes block and Up transition block. The model gains benefits of features reusability due to concatenating path and dense connection because of the dense block. It increased the training speed due to residual networks, and gradient degradation smooth information propagation in the network.

Models with multimodal inputs.

In deep learning, multimodal inputs refer to models that can accept and process multiple types of data simultaneously. This could include a combination of different modalities such as text, images, audio, or other types of structured or unstructured data (Manzoor et al., 2023). Handling multimodal inputs allows the model to leverage information from diverse sources, leading to potentially more accurate predictions or representations. Mao et al., (2023) showed that multimodal (multispectral, thermal, and RGB) data fusion performed better than unimodal data for cold stress detection of tea plants. S. Khan et al. (2019) proposed a DL-based model with multi-modal inputs for the first time where the input data combined processed thermal images, CWSI Index, and Soil Moisture. Wakamori et al. (2020) proposed a multimodal neural network with clustering-based drop (C-Drop) for accurate estimation of plant water stress, using plant image and environmental data. Their proposed method extracts temporal multimodal features from leaf wilting features (physiological data) using environmental features (meteorological data) as an attention mechanism of a multimodal neural network that includes long short-term memory layers. Their result proved the effectiveness of multimodalities in the input features. Zhou et al., (2021a) proposed a disease identification model based on the "image-text" multi modal collaborative representation and knowledge assistance (ITK-Net), which combines image modal features with text modal features and incorporates the knowledge graph for supplementation. In this paper, inference visualization was performed for the three modalities. In the image modality, the model is expected to accurately locate and learn the disease features as far as possible. In the text modality, the disease features were described in human natural language. However, they might be subjected to omissions or errors. In the knowledge modality, the disease knowledge was collected from Baidu Encyclopedia and other related websites, which made the description language highly refined. They indicated that the accuracy of model recognition was improved after adding text modal data compared with a single image input. The disease knowledge graph enhanced the interpretability of the model recognition results. However, the disadvantage of this type of input modality is that the text description increases the time for sample labeling.

Models with multi-scale inputs.

In DL, multi-scale input refers to the use of input data that is processed at multiple scales or resolutions within the same model. This approach can help capture information at different levels of detail, allowing the model to better comprehend the data's underlying structure and context (Elizar et al., 2022). Multi-scale input is commonly used

in computer vision tasks, where objects of interest may vary in size and appearance. Hao et al. (2020) proposed a hierarchical fusion convolutional neural network architecture (MFC-CNN). It is based on multi-scale input that was constructed to grade the light stress. They separated leaf patches from complete leaves and constructed four-scale input so that the local leaf vein and texture information are expanded. In feature extraction, this multi-scale dataset is put into the main network at different depths according to the characteristics of CNN. The architecture has three branches and each branch merges into the master network after a convolutional layer. This operation expands detailed information such as veins and textures by separating multi-patches from intact leaves. This process strengthens the learning ability of detailed information and enables the network to fully extract leaf features from multiple scales.

Multi-task learning models.

Multi-task learning (MTL) in deep learning refers to training a single model to perform multiple tasks simultaneously (Crawshaw, 2020). This approach can be beneficial when the tasks share underlying relationships or when learning one task can help improve performance on another task. Esgario et al. (2020b) proposed a multi-task learning model based on CNN that can classify biotic stress and estimate severity of the stress. The CNN architectures were modified by the addition of a new fully connected layer in parallel with the existing one. In this way, the classification blocks are individualized but the convolutional layers are shared, so that the model learns joint features useful in classifying both problems. The multi-task learning structure reuses most of the network architecture and makes learning substantially faster since only a single model needs to be trained. Salamai & Al-Nami, (2023) developed a multi-task based DL model for visual recognition of biotic stress in coffee leaves.

4. Challenges in plant stress detection based on image and deep learning

Even though DL-based plant stress detection system is promising, it still has some limitations. During image acquisition, environmental elements, including illumination, occlusions, shadows, wind, and background clutter might generate noise and confuse image-based stress detection systems. Furthermore, plants exhibit significant variability in their morphology, color, and growth patterns, even within the same species. Additionally, plant stress symptoms can occasionally take on subtle forms and change based on the type, severity, and length of the stressor. Since some stress reactions can be non-specific or overlap with typical developmental processes, it can be challenging to tell stressed from healthy plants just by looking at their outward appearance. Sometimes multiple stresses occur on a single leaf that can significantly alter the characteristics of symptoms, especially when they overlap. These symptom overlaps are rarely captured correctly, which leads to misclassifications. Hyperspectral data, characterized by a large number of spectral bands, can be computationally intensive and may not always be necessary for stress detection. In contrast, RGB and multispectral images, while computationally efficient, may lack the spectral resolution required to accurately identify stress-related changes in plants. Furthermore, the majority of commercial hyperspectral imaging sensors are currently limited to operating in regulated lighting environments in laboratories, which makes it challenging for data acquisition in field (Z. Gao et al., 2020b).

Image labeling can also be a tedious, labor-intensive, and time-consuming process. Additionally, it could be difficult to obtain precisely labeled data for various stress situations and plant species, requiring expertise in plant disease identification (Dhaka et al., 2021), which could restrict the quantity and quality of training datasets. Model training and evaluation can be skewed by an imbalance between classes of stressed and healthy plants, which can result in poor results, especially for minority classes (Hao et al., 2020). There are tools for generating synthetic images, such as generative adversarial networks (GAN) to mitigate class imbalance issue (Forster et al., 2019). However, the computational overhead to train this model is significant and there is risk of instability, failure of convergence and mode collapse occurrence, where the model is unable to capture the full diversity of the training data distribution, resulting in generated images that lack variation and appear very similar or even identical to each other (GeeksforGeeks, 2024). Moreover, it is challenging to find the right combination of data augmentations for efficient pretraining in different domains. For example, Nagasubramanian et al. (2022) avoided color jittering-based data augmentation during SSL pretraining with soybean disease images to avoid drop in performance.

Most researchers do not consider various factors such as air temperature, soil properties, or leaf area properties while building the stress detection model (Anim-Ayeko et al., 2023a). As a result, DL algorithms may have difficulty generalizing to new plant samples with variable lighting conditions, soil types, or climate considerations if they were trained on image data from particular growth settings or environmental conditions (G C et al., 2022). Moreover, DL models require a large amount of data for training, and data scarcity can reduce the model performance and cause overfitting (S. Khan et al., 2019). CNN models can be computationally demanding and need a significant amount of computing power for both training and inference. The efficiency and scalability of these models deployed on resource-constrained platforms, like edge devices or UAVs, is a further complication. Additionally, the performance of classification models tends to degrade with an increasing number of classes, posing significant challenges in developing lightweight models capable of real-time stress monitoring in field environments. Another challenging factor is choosing the optimal model parameters, such as learning rate and epochs can be difficult. In most cases, the apps developed for plant stress detection would need internet access for its operation, which might not always be available to the user (Esgario et al., 2022).

5. Future direction

In order to build a robust model that has more generalizing ability to new plant samples, it needs to be trained with plant images collected in more realistic background conditions, such as overlapping plants, changes in illumination conditions, or other situations that might appear in real-life (Gómez-Zamanillo et al., 2023). Moreover, the training data should include plant samples from different growth settings or environmental conditions, such as soil type, climate, and lighting condition. Also, the model can be trained with multi-modal data from a variety of imaging platforms (e.g., multispectral, hyperspectral) and on-field protocols (UAVs, ground imaging, satellite) and various growth stages to improve robustness. Other data modalities, such as temperature, humidity, time information, growth stage, and genetic data can be integrated with the image data to improve the model performance. Since the field sample collection is time- consuming and labor-intensive, image-based fast and automatic training sample collection methods should be developed to handle this issue (Dao et al., 2021a).

More research needs to be done to efficiently reduce the computational overhead of preprocessing techniques in DL pipeline, such as noise reduction and segmentation. Furthermore, to mitigate the class imbalance issue, class balancing techniques, such as under-sampling, over-sampling, or augmenting the minority class could be integrated to the DL pipeline. Creating tools for accurate and efficient image labeling can reduce the tedious process of manual image labeling. Furthermore, automated tools for creating customized preprocessing pipelines based on specific crop types, environments, and stressors may improve model performance.

Current studies are working on improving the data limitation issues by implementing data augmentation, transfer learning, and semi-supervised learning techniques (e.g., GAN). Another crucial area to focus on is model light-weighting, which will enable models to be faster and more easily integrated on small devices, like drones. At present,

there are two main directions for obtaining lightweight neural network models: (1) artificially designed lightweight neural networks such as MobileNet and Xception, (2) automatically designed lightweight neural network models based on neural network architecture searches such as EfficientNet, and CNN model compression (Chen et al., 2023). DL models are often regarded as black-box systems, making it challenging to interpret their decisions and understand the features driving classification outcomes. Ensuring model interpretability and explainability is important for gaining insights into plant stress mechanisms and building trust in model predictions. One way to alleviate the problem of multiple stress indication on a single leaf would be to train a similarity-based architecture that can rule out those symptoms that are not similar to any of the stresses in the dataset by classifying them into a new class, for example the *others* class (Esgario et al., 2022).

6. Conclusions

Deep learning-based plant stress detection algorithms have demonstrated great promise in recent years, providing novel approaches to recognizing and managing stressors in agricultural contexts. Recent improvements in sensors have enabled the collection of massive amounts of high-quality and high-resolution imaging data, such as RGB, multispectral, hyperspectral, and so on. Large-scale monitoring of plant health and stress levels has been made possible by the integration of DL models with remote sensing platforms, such as drones fitted with multispectral or hyperspectral cameras. The advancement of DL has transformed image-based tasks, such as stress classification, disease detection, and segmentation. The accessibility of frameworks like TensorFlow and PyTorch has fueled their widespread adoption across domains, including plant stress detection, where various DL architectures like VGG16, ResNet, Inception, EfficientNet, and MobileNet are commonly utilized. Furthermore, the advancements in TL and FSL techniques for handling data scarcity, the application of multimodal data integration to improve model robustness, and the role of transformer-based models in capturing complex dependencies in image data have significantly advanced the practicality of plant stress detection systems. Despite their success, DL-based plant stress detection systems encounter some limitations such as quality data scarcity, data labeling, model generalizability and domain adaptability. Subsequent investigations on these limits can concentrate on improving data augmentation, transfer learning, and model light-weighting strategies. The full benefits of these techniques in agricultural practice will only be realized with continued progress in model development, data gathering, and implementation strategies.

CRediT authorship contribution statement

Nijhum Paul: Writing – original draft, Validation, Methodology, Data curation. **G.C. Sunil:** Writing – review & editing, Validation, Methodology. **David Horvath:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation. **Xin Sun:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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